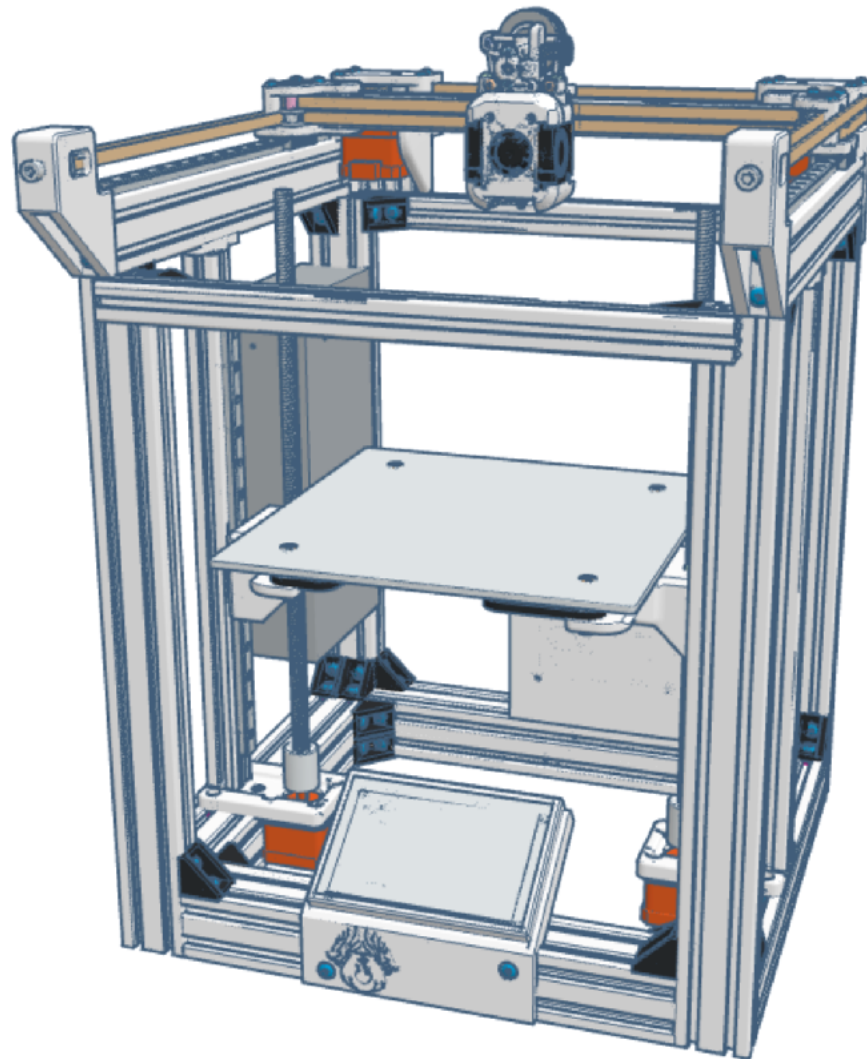




Duender



Your Open-Source 2 x Creality Ender-3 core XY conversion

by Sergei Irbis

Some points of the build require your attention



Once the printer is complete, the print bed and the nozzle will be heating and may cause **severe burning** if you are not cautious.
Do not touch those parts while in function.



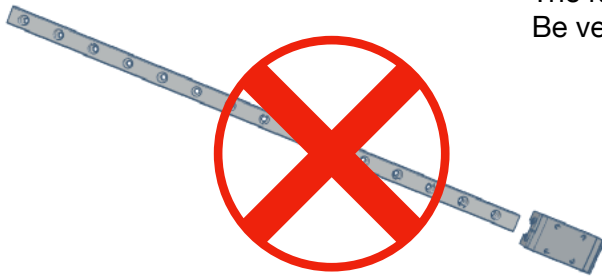
A risk of overheating or electric short in components is always possible. In some cases, this could lead to the printer **starting a fire**.

Please, double check your build and babysit your printer during the tuning phase as well as its first jobs

Keeping an **extinguisher** near those appliances is a good practice.

Read the full doc before starting your build, and do not hesitate to reach out on the [Discord server](#) to ask for help any time you are unsure

Linear rails handling



The rail axis carriages are meant to slide smoothly with very little resistance.
Be very careful **they do not fall off the rail**, as this would most likely result in irreparable damage.

The MGN9H rails are quite thin and can be somewhat difficult to fit onto V-slot extrusions.
You can use some **3D printed helpers** to align them: use [spacers](#) or tools like [this](#) or [this](#).
Try tightening the screws progressively along the rail while maintaining alignment until all screws are secured.

When you receive them, they are very often covered in some kind of protective oil.
The **video below by Canuck Creator** offers all the advice you need to clean and lubricate your rails.



<https://2cm.es/1ml5J>

Table of contents

Hardware reference	1
Bill of materials	2
Important tips	4
Assembling the frame	6
Z gantry	17
AB gantry	22
X axis gantry	29
Dragonburner toolhead	34
Power supply, control box and screen housing	45
Wiring	48
A few words about Medusa HC project	55

Need more information?



<https://2cm.es/1h2CJ>



<https://2cm.es/1h2CE>



<https://2cm.es/1IYzA>

Wanna support the project?

If you enjoy Duender and want to support what I do, there are a few ways to help:

Patreon — monthly support

<https://patreon.com/Irbis3D>

Ko-fi / Buy Me a Coffee — one-time donations

<https://buymeacoffee.com/Irbis3D>

YouTube Super Thanks — under any video

<https://youtube.com/@Irbis3D>

Your support helps me create more content, upgrade gear, and keep experimenting with cool ideas.

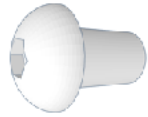
Thank you.

Sergei Irbis.

Know your bits

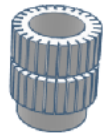
Hardware reference

1



BHCS (Button Head Cap Screw)

Metric fastener with hemispheric head and hex drive. Mostly used with M5 T-slot nuts



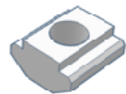
Heat set insert

Once heated with the soldering iron, it melts into plastic, which makes a tight fit by solidifying.
M3 threading, 5mm outer diam., 4mm long.



FHCS (Flat Head Cap Screw)

Metric fastener with cone shaped head and flat top



T-slot nut

Nut that is inserted inside the slot of aluminium extrusions. Must be inserted on beforehand
Hammerhead T nuts can be used after assembly



Hex nut

Coupled with a screw to create a bolt.
Nylstop ones have a plastic insert to prevent them un-tightening



SHCS (Socket Head Cap Screw)

Metric fastener with cylindrical head and hex drive. Mostly used with T-slot nuts.



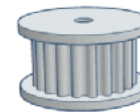
Shim (or spacer)

Metal flat ring, with a set thickness, used as spacer between parts and bearings. Not to be confused with washers



Pin

Smooth precision-cut metal rod used as an idler axle.



Toothed idler

Has bearings in the center bore, used to redirect the belt in the motion system. GT2 refers to the belt pitch between teeth



Toothless idler

Similar idler without teeth, for the smooth side of the belt. 16T is for teeth count, 6W for belt width, 3B for axis bore diameter

BOM and sourcing

Bill of materials

2

Fasteners	Qty	Order	Affiliated link / Notes	Kinematics	Qty	Order	Affiliated link / Notes
M5x25 SHCS	2	✓	From Ender 3	42-34 Nema17 motor	2	✓	From Ender 3
M4X35 FHCS	4	✓	From Ender 3	42-40 Nema17 motor	2	✓	From Ender 3
M4x20 SHCS	2	✓	From Ender 3	Leadscrew T8x8mm 365mm	2	✓	From Ender 3
M4 hex nut	4	✓	From Ender 3	Motor coupler D5x8 <i>Optional elastic coupler</i>	2	✓	From Ender 3
Metal corner bracket	28		https://s.click.aliexpress.com/e/_c4DUW5sX		2		
M5x8 BHCS	92			T8 Lead screw	2	✓	From Ender 3
M5 T-slot nut	79		https://s.click.aliexpress.com/e/_c4EDoXzD	Leadscrew brass nut <i>Optional anti backlash nut</i>	2	✓	From Ender 3
M5 hex nut	2				2		
M3x30 BHCS	14			Y plate	1	✓	From Ender 3
M3x20 BHCS	4			10t Nema 14 motor	1		https://s.click.aliexpress.com/e/_c4NfNSaf
M3x18 BHCS	8			MGN9H or MGN12H 300mm rail	5		https://s.click.aliexpress.com/e/_oFYW0St
M3x16 BHCS	2			GT2 6mm wide belt	5m		https://s.click.aliexpress.com/e/_ooDLTEH
M3x14 BHCS	2			Pulley GT2 16T 6W 5B	2		https://s.click.aliexpress.com/e/_c4mjCdxl
M3x12 BHCS	9			Toothless idler GT2 16T6W3B	10		https://s.click.aliexpress.com/e/_c3Fqpe71
M3x8 BHCS	74			Toothed idler GT2 16T6W3B	2		https://s.click.aliexpress.com/e/_c3Fqpe71
M3x6 BHCS	24			M3x20mm pin	3		https://s.click.aliexpress.com/e/_c4CXA4ZN
M3 T-slot nut	40		https://s.click.aliexpress.com/e/_c4EDoXzD	M3x30mm Pin	8		https://s.click.aliexpress.com/e/_c4CXA4ZN
M3 hex nut	13			M3x0.5mm shim	38		https://s.click.aliexpress.com/e/_c4BNsuvz
M3 heat set insert	44		https://s.click.aliexpress.com/e/_c4DbNkaX	M3x10mm spacer	6		https://s.click.aliexpress.com/e/_c3rUz3wF

Long screws kit: https://s.click.aliexpress.com/e/_c4VYXcQR (colour: Button head)

Short screws kit: https://s.click.aliexpress.com/e/_c30o44Xt

Electronics	Qty	Order	Affiliated link / Notes
24V 15A PSU, housing, switch, cable	1	✓	From Ender 3
Heatbed 24V 235x235mm	1	✓	From Ender 3
4010 24V axial fan	2	✓	From Ender 3
BTT HDMI5 V1.2	1		https://s.click.aliexpress.com/e/_c4odBUeJ
BTT Manta 5P BTT Manta 8P V2.0	1 1		https://s.click.aliexpress.com/e/_oktZaKt
BTT CB2	1		https://s.click.aliexpress.com/e/_oDMokmp
BTT TMC 2209	5		https://s.click.aliexpress.com/e/_oktZaKt
Misc	Qty	Order	Affiliated link / Notes
2020 V-slot extrusion 300mm	1		Unless you plan for Medusa HC, see p.16 first
12mm silicone bed spacer Optional Spring bed spacer	4 4	✓	From Ender 3
Bambulab X1/PIP hotend	1		https://s.click.aliexpress.com/e/_c3yvXGSr
3010 24V axial fan	1		https://s.click.aliexpress.com/e/_c3YNV2ux
4010 24V radial fan	2		https://s.click.aliexpress.com/e/_c3eDwMjh
Sherpa mini Thumbscrew w/Spring	1		https://s.click.aliexpress.com/e/_c4kf7IXN
Sherpa mini filament gear	1		
Sherpa mini idler gear	1		
Wires, connectors, pins, crimping pliers and stuff			

Are you planning for Medusa HC?

Medusa HC is a toolhead changer specifically designed for the Duender offering by default 4 swappable hotends.

If you use a different control board, with for example a Raspberry Pi as your Klipper host, you may have to add a second board to get enough hotend heaters and thermistors ports available.

However, if you are considering using the BTT Manta 5P recommended here, and planning for Medusa HC, it is advised to switch it right away for a **BTT Manta 8P V2.0** with a CB2 card and the TMC2209 drivers so you can expand and connect the hotends with no added modification.

BTT Manta 8P V2.0 wiring is detailed in the Medusa HC guide

Duender print settings:

Material: ABS or ASA for all load-bearing parts

Electronics box and display holder can be printed in any material

Layer height: 0.2mm

No. of walls/top/bottom layers: 5

Infill: > 50% (60% recommended) for load bearing parts
Gyroid for strength, **cross-hatch** for speed.
 Can be as low as 15% for electronics or screen boxes

Last minute call

Important tips

Ender 3 disassembly

DO NOT DISMANTLE THE POWER SUPPLY. You will use the PSU along with its cover, the power switch and the wire ending with an XT60 plug.

Deconstruction is easier if you first remove all connections from the mainboard. Those are held with hot glue, carefully remove it with a pair of pliers. You can pull out the X axis after removing the top bar, which simplifies deconstruction. The X motor has a QR code sticker hiding four screws. The 4040 central profile and plastic parts won't be needed. Remove the rubber feet for easier squaring, but you may want to put them back later.

Depending on your revision, you may have to **take the Ender 3 motors pulleys off**. Some use a [printed tool](#), some just Dremel them out. Be careful!

Linear rails

This manual depicts MGN9H build on the images, however, it also applies to MGNH12 builds as the steps, hardware and procedures are the same. Be careful however and make sure you do print the right parts for your build.

Z Axis

It is probably a good idea, to replace the Z motors shaft couplers with "spring motor coupler" ones at approx. 2x€2.50. In the same domain, it is advised to replace original brass lead screw nuts with anti-backlash spring loaded ones at approx. 2x€2.50 too.

Toolhead

The toolhead of choice is a **Dragonburner** with a **Bambulab X1/P1P series hotend**, associated with a **Sherpa mini** extruder. This will allow for easier switching to Medusa HC should you want to: your carriage, fans, hotend and most extruder parts can then be reused. If you want to go for Medusa HC right away, switch to the relevant manual, and skip the toolhead assembly mentioned here.

Electronics

if you want to be really frugal, any Klipper board will work, however, the **BTT Manta 5P with a CB1/2** is likely the best choice for your Duender. Note that if you plan on building Medusa HC at some point, you would be better off starting with a **BTT Manta 8P+CB2** from the beginning, as it offers all the necessary ports for your Duender with Medusa HC attached.

Original design goes with a BTT HDMI5. You may not need a display, as you'll be able to use the web interface anyway, but the original 12864 of your Ender 3 can do the job, or you can get any other Klipper compatible touch screen (*you may need to tweak the config files though*).

Remember that you might need different hardware if you made different electronics choices.

Last minute call

5

Building

During the frame assembly, **it is crucial that everything ends up square and flush**. The base and four vertical posts should be placed first then squared while gradually tightening the fasteners. It is tedious but the print quality of your Duender depends on it.

Since MGN9H rails are about as wide as the V-slots, it's not easy to get them centred. You can [print some spacers](#) or alignment tool like [this](#) or [this](#).

The belt tensioners are also quite fiddly since you need to get all washers and idlers lined up before putting the pin in. [A printed jig](#) may help you.

If you are unsure about how to correctly tap your extrusions, check out this video first:

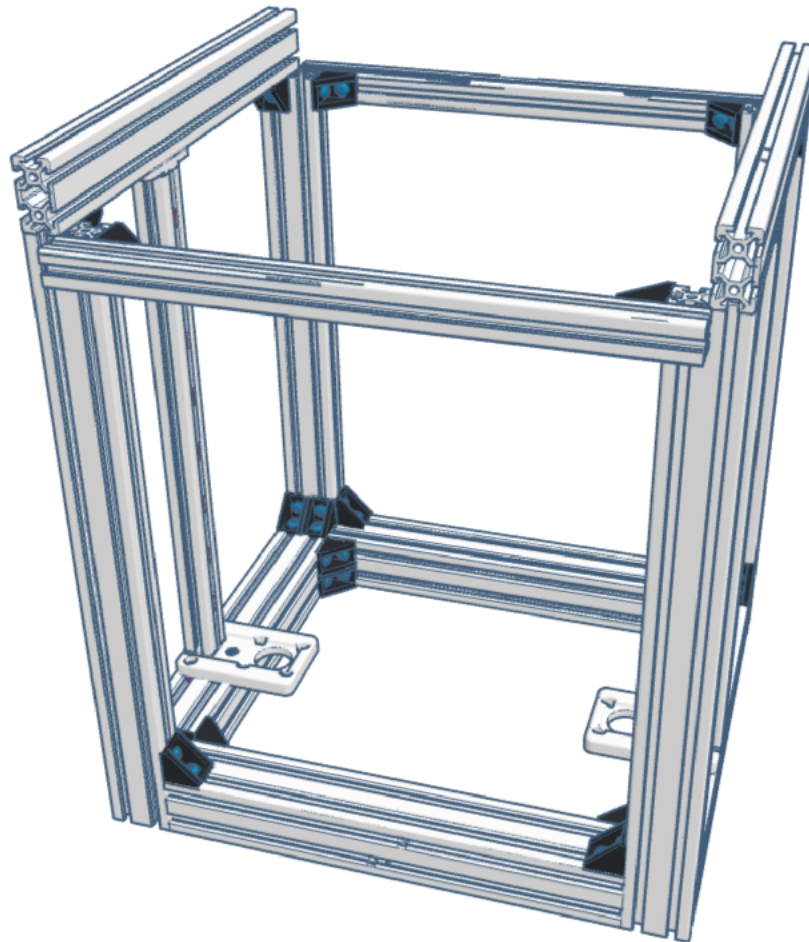


<https://2cm.es/1YBi>

Base structure

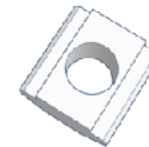
Assembling the frame

6



Metal corner bracket

28



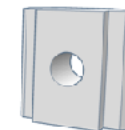
M5 T-slot nut

71



M5x8 BHCS screw

70

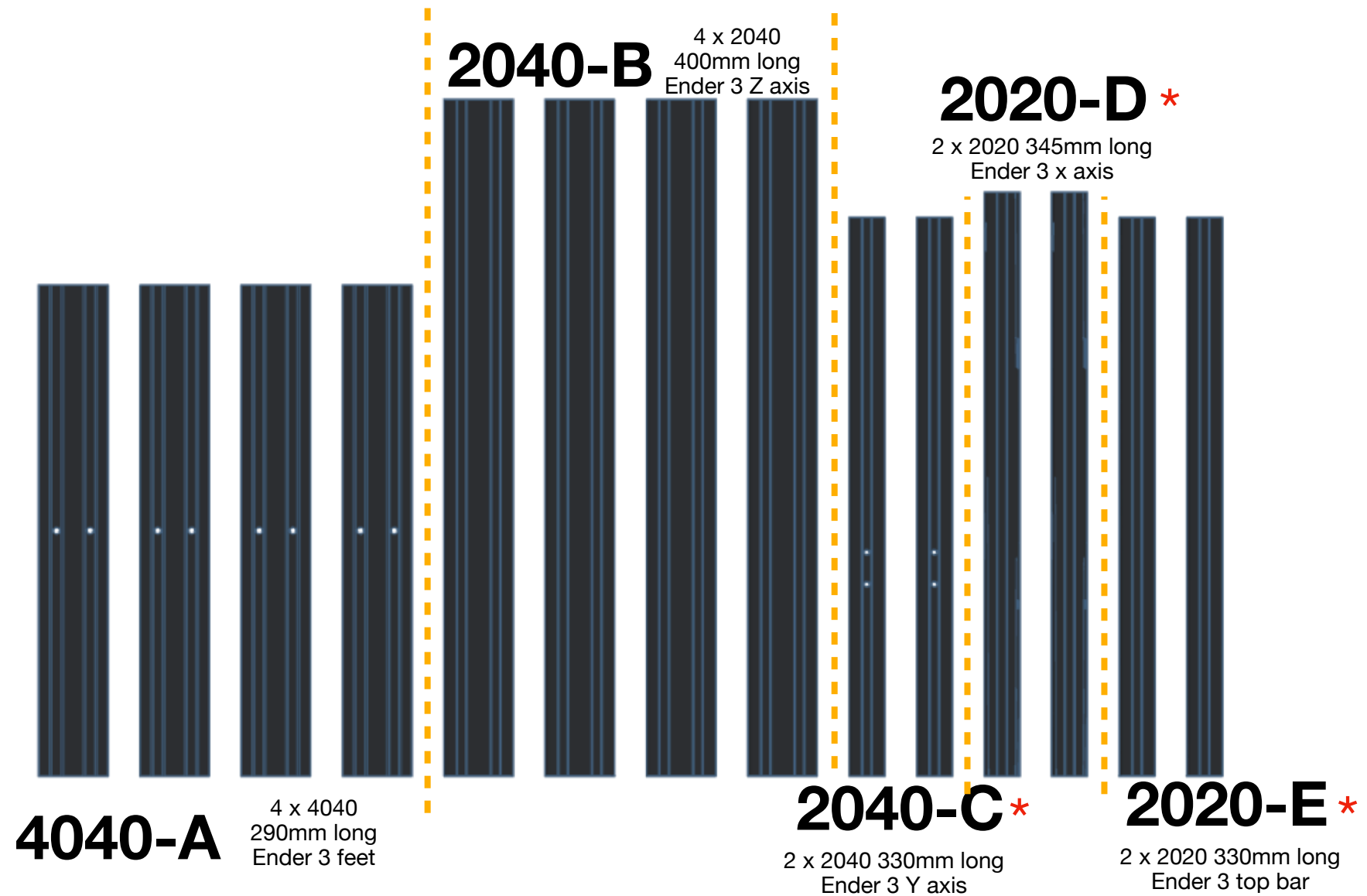


M3 T-slot nut

16

Base structure

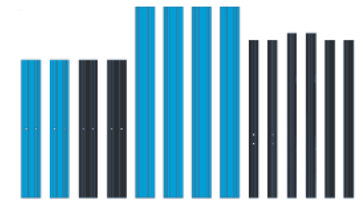
7



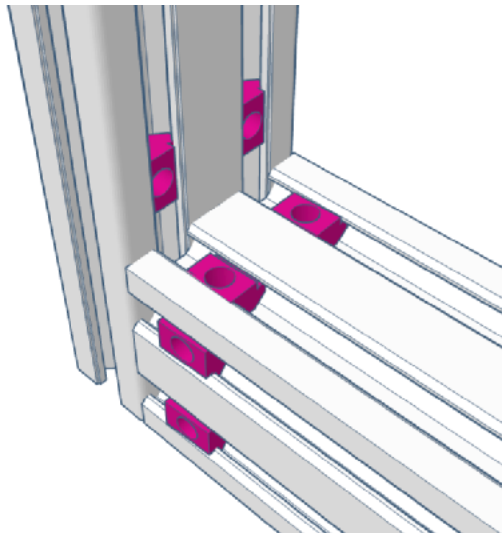
* Extrusions marked with a * need to be M5 tapped at both ends: 2040-C (8 channels), 2020-D (4 channels) and the back 2020-E (2 channels).

Base structure

8

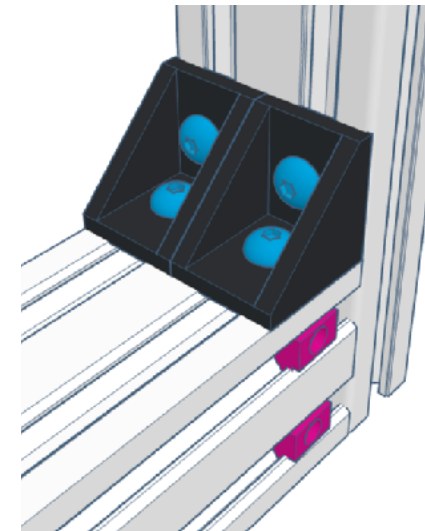


First, prepare each 4040-A **side** with 11 M5 T-slot nut as shown here
You can keep the original Ender 3 plastic pads as feet on the underside
(22 M5 T-slot nuts total)



Insert 2 M5 T-slot nuts in both the 2040-B
(8 M5 T-slot nut total)

**Make sure to orient the back 2040B
extrusions so the mounting holes for the
power supply are towards the inside of
the printer (see picture on next page)**



Place 2 brackets and
insert the M5x8 screws

Repeat the process at
the opposite end of
each 4040-A
(16 M5x8 BHCS total)

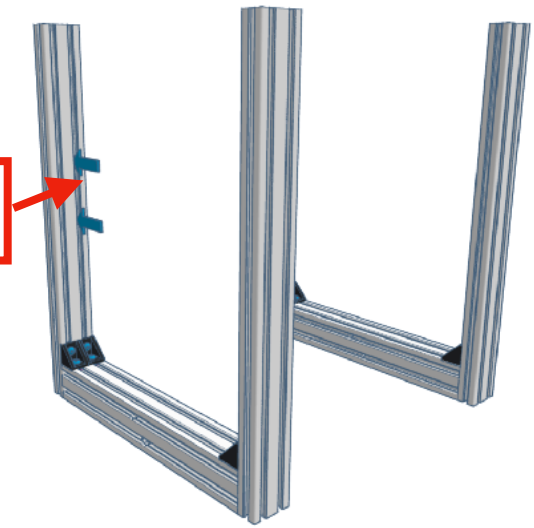
Base structure

9

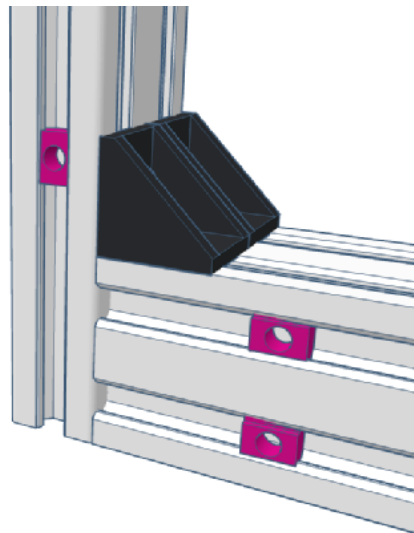


Make sure everything is flush and square before final tightening

Make sure the holes for the power supply end on the inside of the back 2040-B



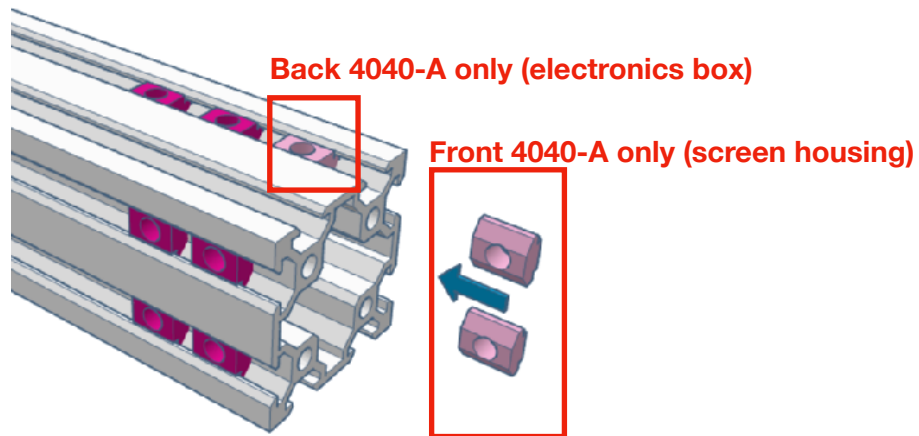
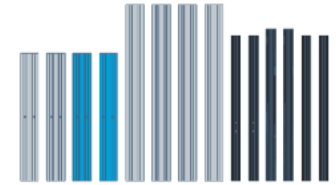
Once done, repeat the same steps with the remaining 4040-A and the two 2040-B to form the opposite side of the frame



Insert a M5 T-slot nut in the inside slot of each 2040-B post of the previous assemblies (4 M5 T-slot nuts total)

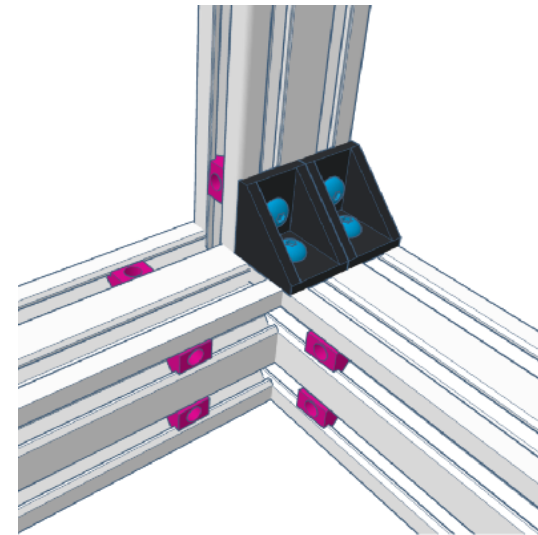
Base structure

10

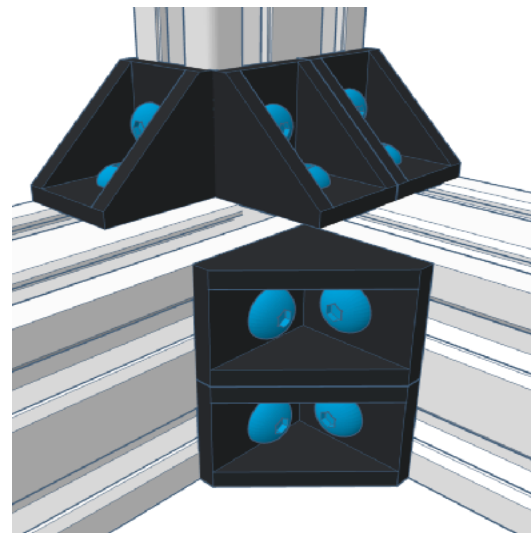


Insert 6 M5 T-slot nut in each of the remaining 4040-A part as shown here (12 total)

Add the M5 T-slot nuts shown in picture in the relevant extrusions (so 15 total)



Place the 4040-A flush and square to one side assembly and proceed installing the brackets with 6 M5x8 screws on each end (24 M5x8 BHCS total)



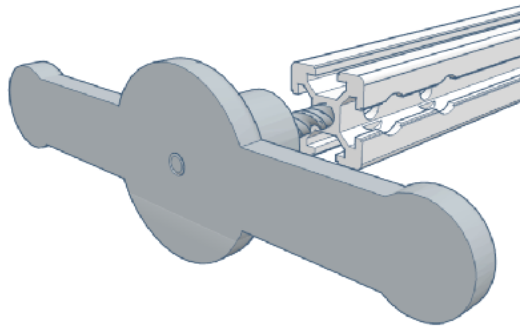
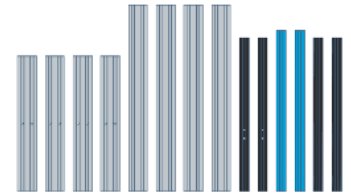
Repeat those steps in the four corners of the base

Once done, double check all your alignments

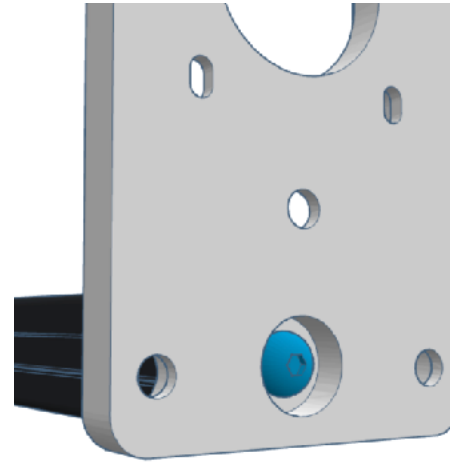
It's a good idea to measure each diagonal of the base square to make sure they are equal too!

Base structure

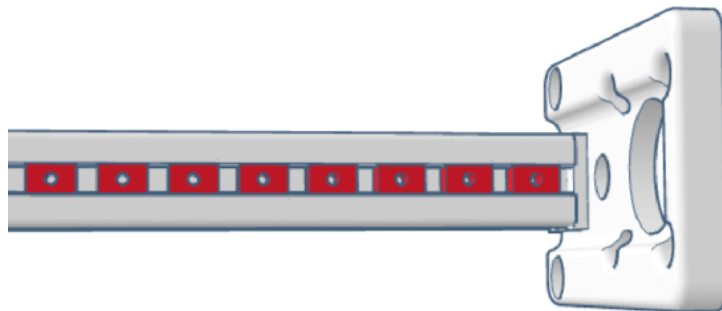
11



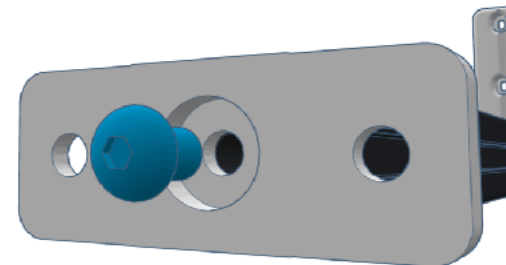
Carefully tap each end of the channel in both the 2020-D extrusions with an M5 bit



Install the Z motor mount onto one end of the extrusion, making sure it's seated and flush, then secure it with a single M5x8 BHCS screw
(2 M5x8 BHCS total)



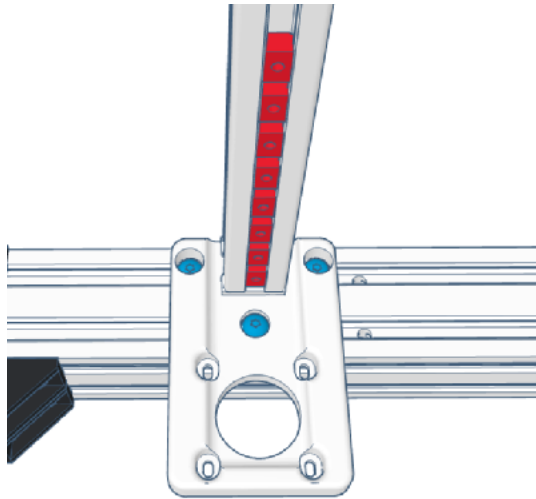
Insert 8 M3 T-slot nuts from the top, facing the same direction the motor is to be mounted.
(16 M3 T-slot nuts total)



Do the same thing with the Z axis top spacer, as pictured
(2 M5x8 BHCS total)

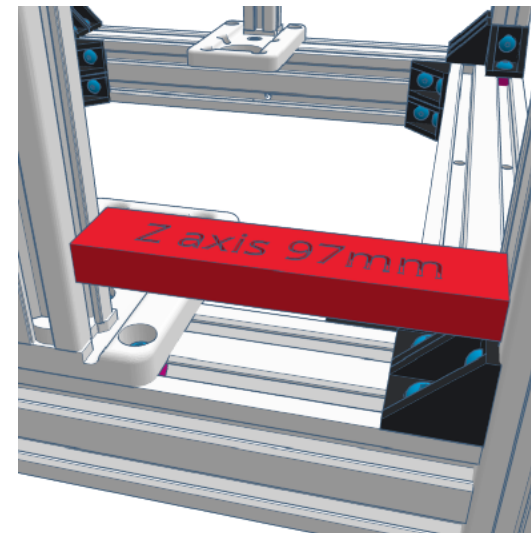
Base structure

12



Install the Z axis onto one side of the base
Align the previously installed M5 T-slot nuts with the holes
Secure it with three M5x8 BHCS screws
(6 M5x8 BHCS total)

DO NOT FULLY TIGHTEN THE SCREWS YET
The assembly must remain free to slide side-to-side for later adjustments.

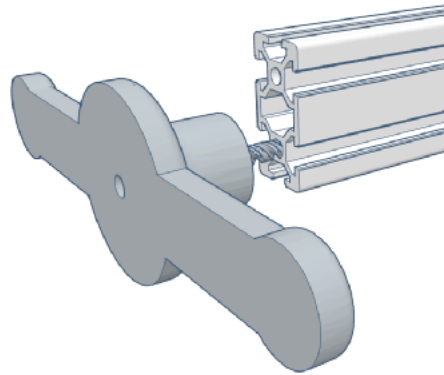
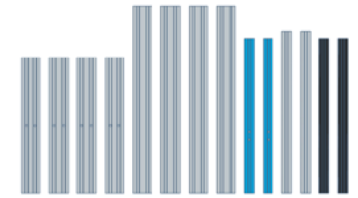


You can 3D print a 97mm spacer, or use a calliper to adjust the Z axis spacing from the front post, then tighten the 3 M5x8 BHCS screws of the base

This is the default distance. You may need to adjust this distance later, depending on your toolhead configuration

Base structure

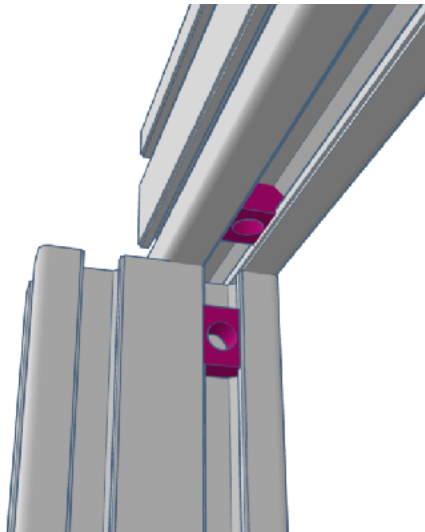
13



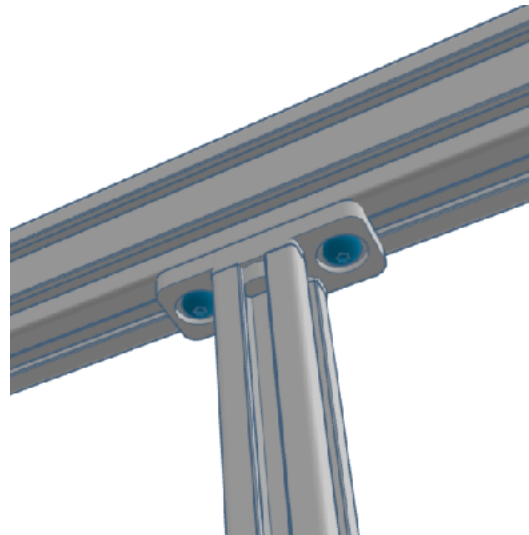
Check whether the channels of your 2040-C extrusions are already tapped. If not, carefully tap each one on each end of both extrusions with an M5 bit (8 channels total).



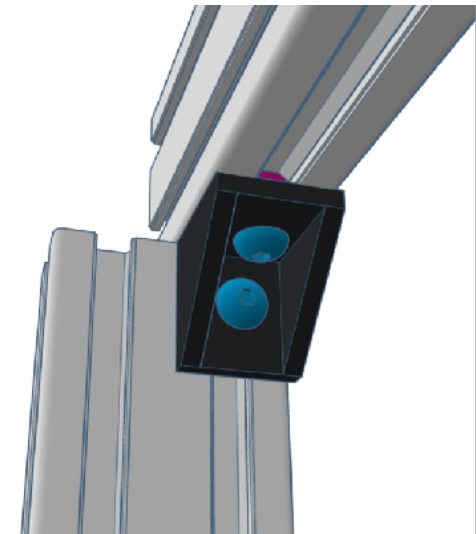
Once done, insert 4 M5 T-slot nuts on the underside of each one (8 M5 T-slot nuts total).



Insert an M5 T-slot nut on the top of each post, in the outside rail as shown (4 M5 T-slot nut total).
Place the 2040-C on top of each side.



Align the 2 M5 T-slot nuts remaining under the 2040-C with the top Z axis plate, and use 2 M5x8 BHCS screws to secure it (4 M5x8 BHCS total).
Check the distance to the front post to be 97mm all along the Z axis before tightening the screws.

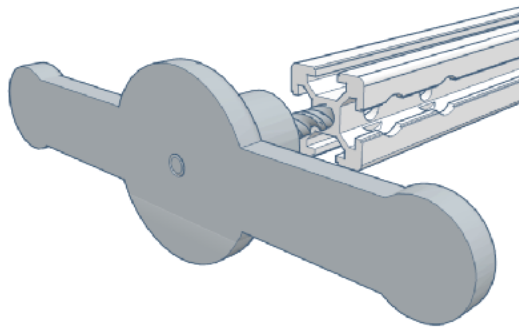
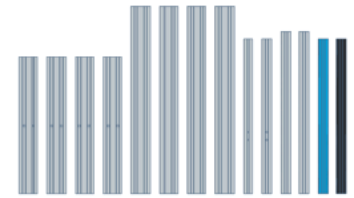


Make sure the extrusions are flush at each end, then proceed and install the corner brackets.

Repeat those steps on the other side to assemble the second Y axis to the frame (8 M5x8 BHCS total).

Base structure

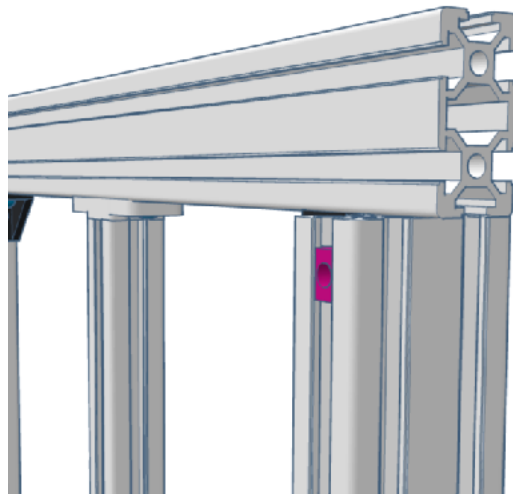
14



Carefully tap each end of the channel in the back 2020-E extrusion with an M5 bit

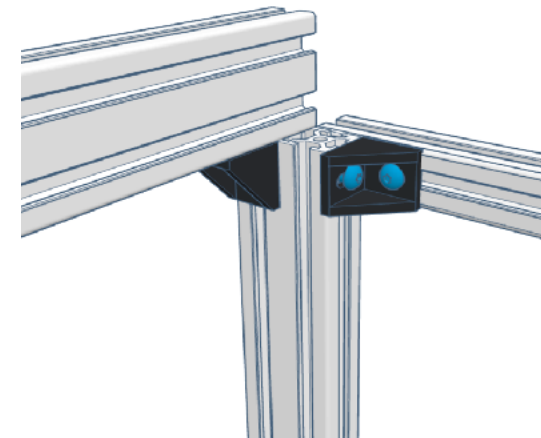


Once done, insert 2 M5 T-slot nuts into the back rail, and 2 more into the front one (4 M5 T-slot nuts total)

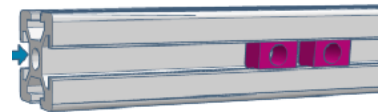
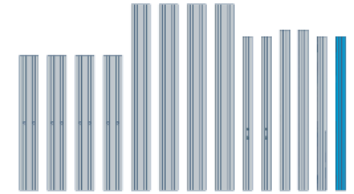


Insert one M5 T-slot nut at the top of each rear Z posts (2 M5 T-slot nut total)

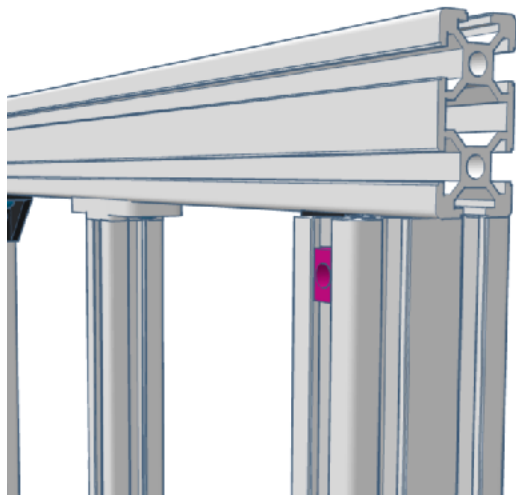
For the rear right Z post only: add a second M5 T-slot nut (for the electronics box later on)



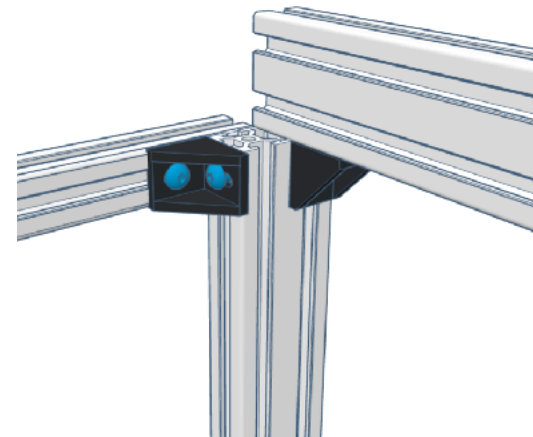
Align the 2020-E extrusion flush to the top of the Z posts
The 2020-E must be centred between the posts
Align the T-slot nuts
Install a corner bracket with 2 M5x8 BHCS screws (4 M5x8 BHCS total)



For the front brace, insert 2 M5 T-slot nuts into the back rail



Insert one M5 T-slot nut at the top of each front Z posts
(2 M5 T-slot nut total)

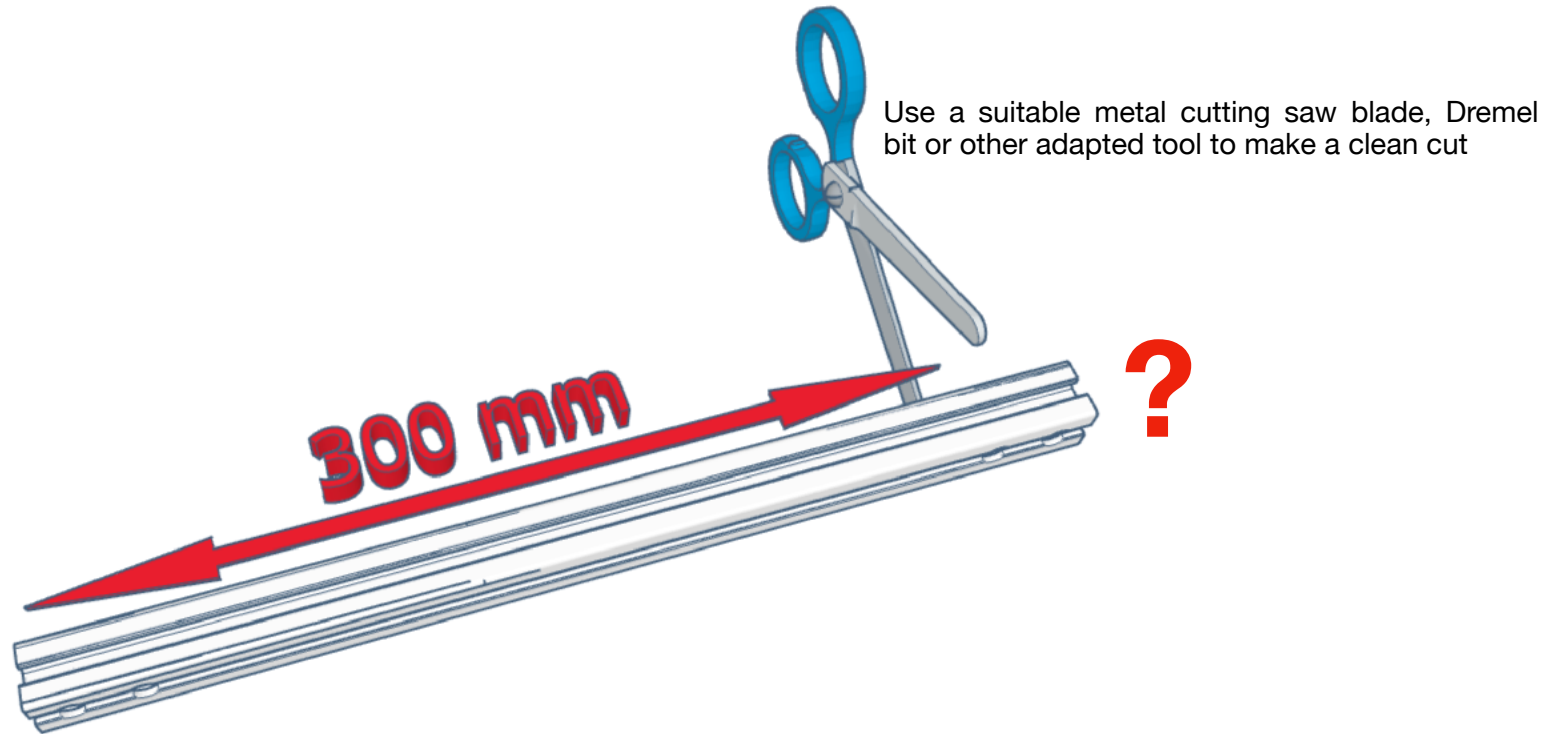


Align the last 2020-E extrusion flush to the top of the Z front posts

The 2020-E must be centred between the posts

Align the T-slot nuts

Install a corner bracket with 2 M5x8 BHCS screws
(4 M5x8 BHCS total)



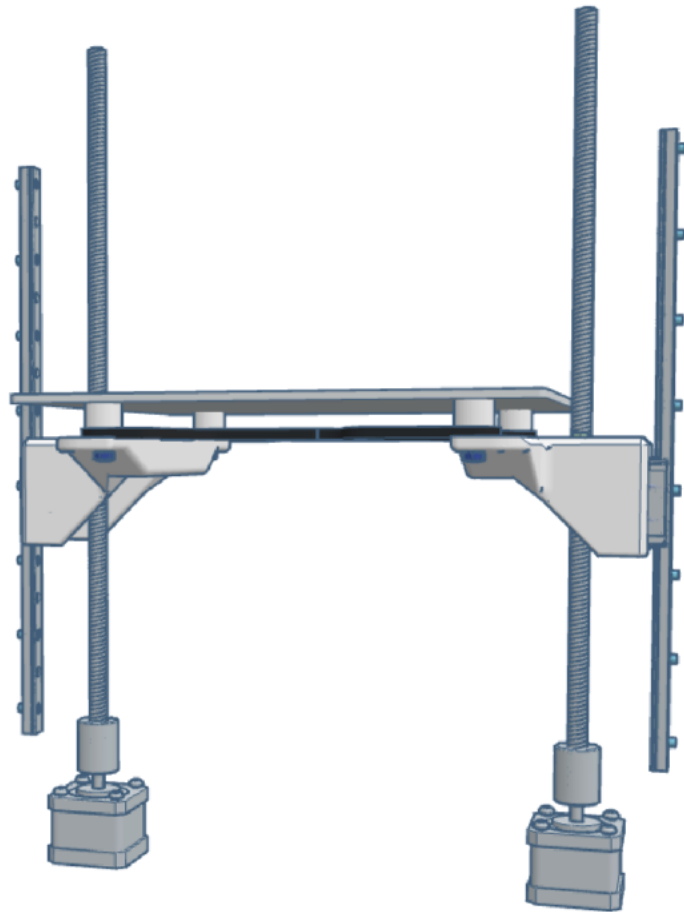
The freshly bought 300 mm long extrusion is to be used to build the X axis, a few sections later.

NOTE that if you are planning for MedusaHC mod later, the front brace needs to be 350mm long.

So the best solution is probably to cut the 2020-E front brace to 300mm and use it for your X axis, and buy directly the 350mm 2020 extrusion for the Medusa HC front brace (or buy both 350 and 300mm if you prefer not to cut your extrusions).

Assembling the Z axis Z gantry

17



M3x12 BHCS screw

4



M3x8 BHCS screw

32



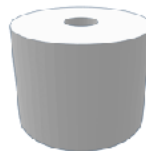
M4 Hex Nylstop nut

4



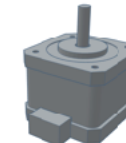
M3 Hex Nylstop nut

4



12mm silicone spacers

4



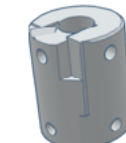
Nema 17 42-34
(from Ender 3)

2



M4x35 FHCS
(from Ender 3)

4



Motor shaft coupler
(from Ender 3)

2



8mm lead screw
(from Ender 3)

2

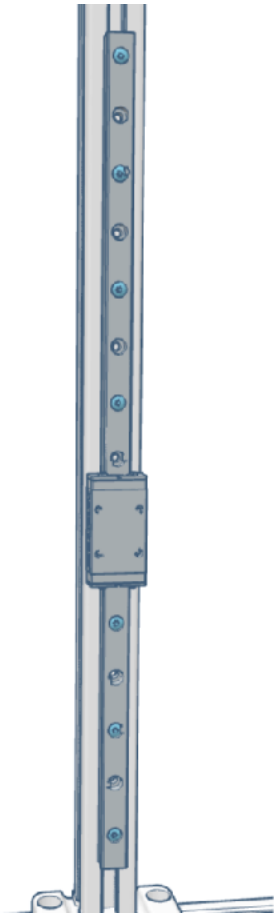


Lead screw nut
(from Ender 3)

2

Assembling the Z axis

18

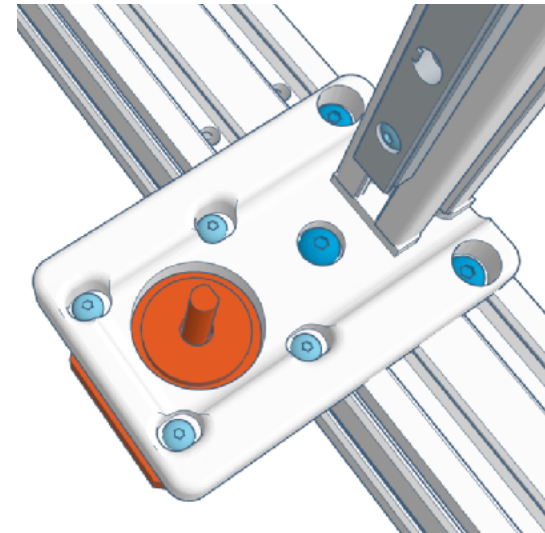


Start by placing one linear rail, MGN9H or MGN12H, on both side Z axis and secure it with 8 M3x8 BHCS every other hole (16 M3x8 BHCS total).

You should use a plastic jig to center the rail on the extrusion especially if you are using MGN9H rails.

Base of the rail is at 25mm distance from the bottom of the extrusion.

Take extra precaution so the carriage does not fall off the rail: place plastic plugs, use tape, whatever works for you.

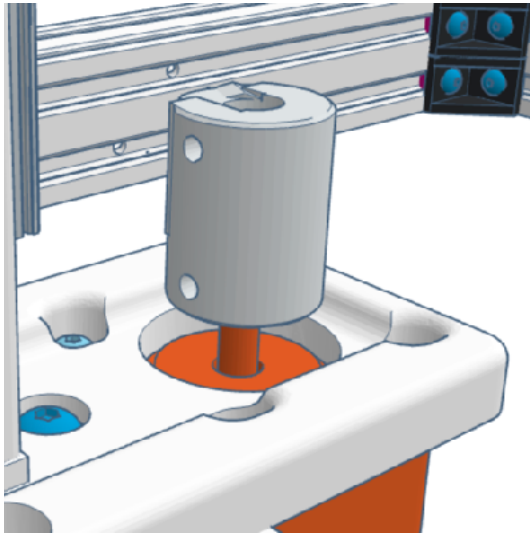


Place the Z motors under their plates and use 4 M3x8 BHCS screws to secure them (8 M3x8 BHCS total)

Orient the motor plug to the back of the printer

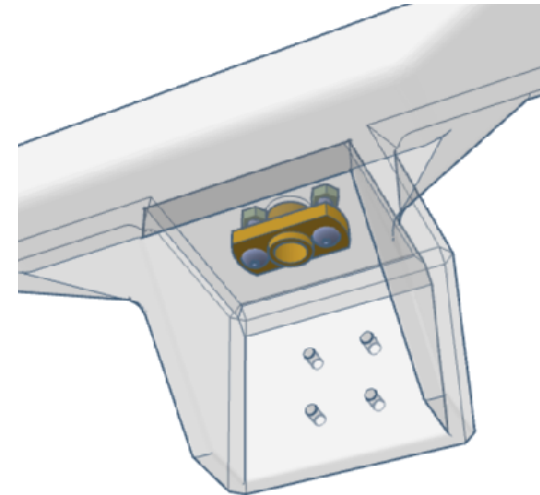
Assembling the Z axis

19



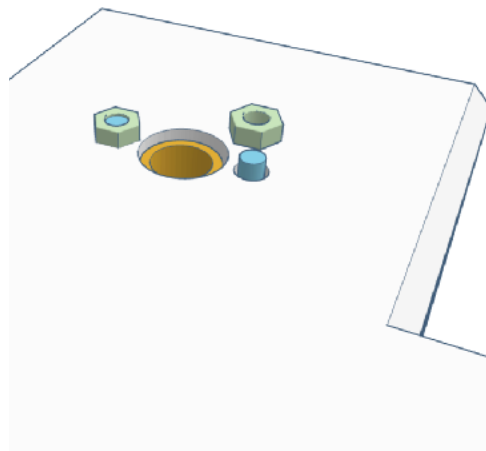
Install the lead screws couplers onto the Z motors shafts and tighten only the lower screw

If your lead screws aren't perfectly straight, you may want to replace those with "spring" or "elastic" ones to avoid Z-wobble



Install the lead screw nuts under the plastic bed holders, and insert 2 M3x12 BHCS screws (you can also use the ones originally coming from your Ender 3) (4 M3x12 BHCS total)

If your lead screws aren't perfectly straight, it is advised you replace them with some "anti-backlash" ones, to avoid Z wobble

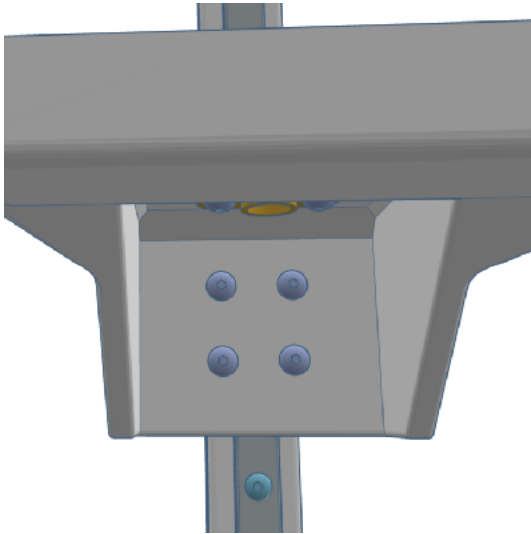


Turn the bed holders around being careful not to lose the screws and secure them with M3 nuts. (4 M3 Hex nut total)

Alternatively, you could insert 2 M3 heat-set inserts (not accounted in the bom) from the top, and tighten the M3 screws from bottom.

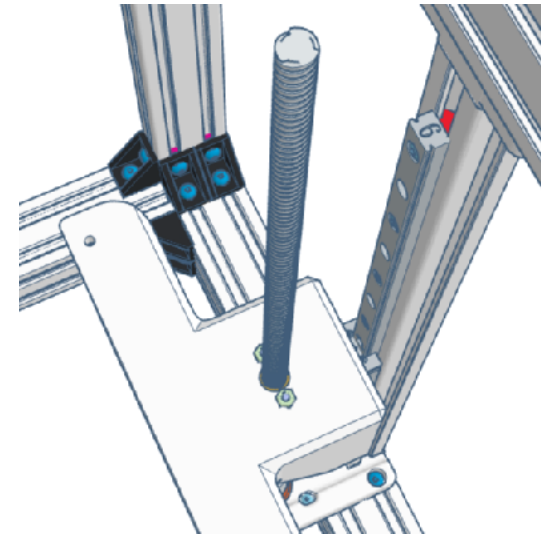
Assembling the Z axis

20

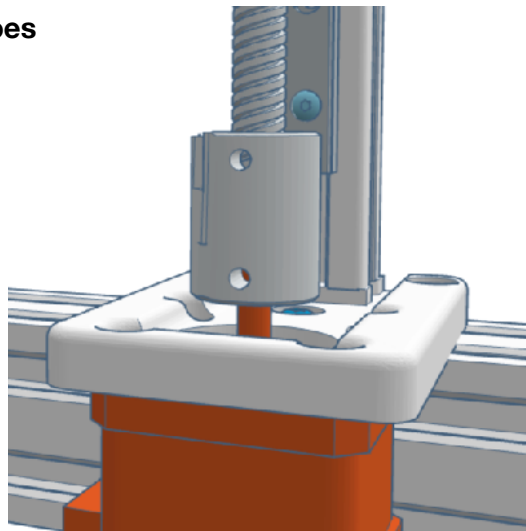


Place each bed holder onto its rail carriage and use 4 M3x8 BHCS screws to attach it (8 M3x8 BHCS total)

Mind the orientation: the longer arm goes towards the front of the printer.



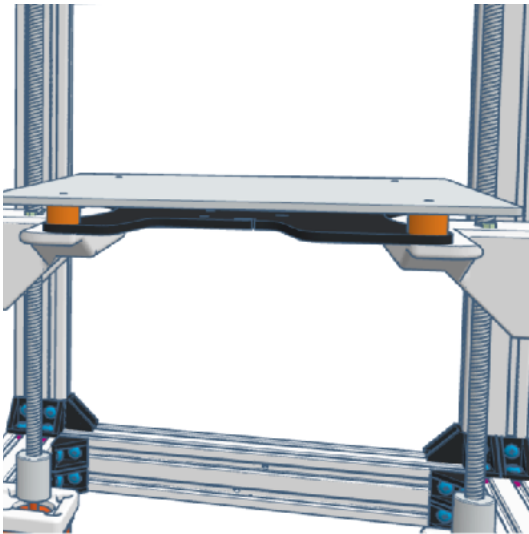
Insert the lead screws into the lead screws nut from the top then rotate them until fully seated into the couplers



Tighten the upper screw of the coupler to hold the lead screw

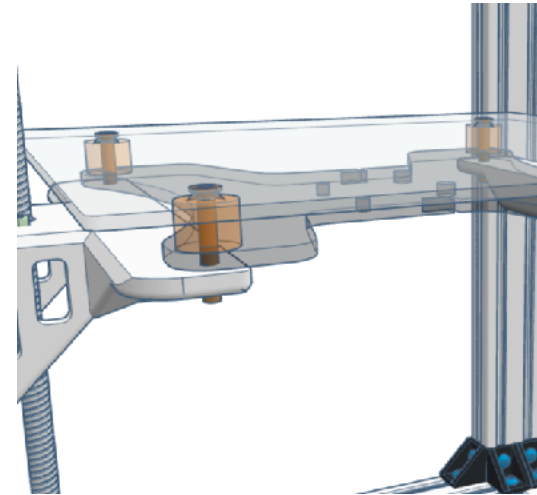
Assembling the Z axis

21

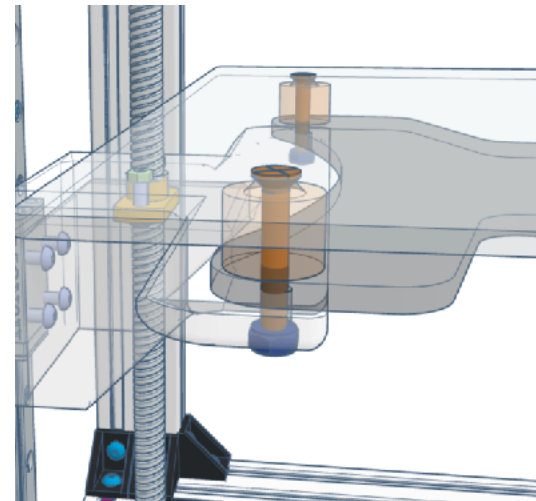


Place the Ender 3 Y plate onto the bed holders, 4 silicone spacers -here 12mm shown in orange (you may use the springs from the Ender 3) then the heatbed

Of course, the wiring of the heatbed should end towards the back of your printer



Use the original 4 M4x35 FHCS screws and secure them under the bed holders with M4 nuts

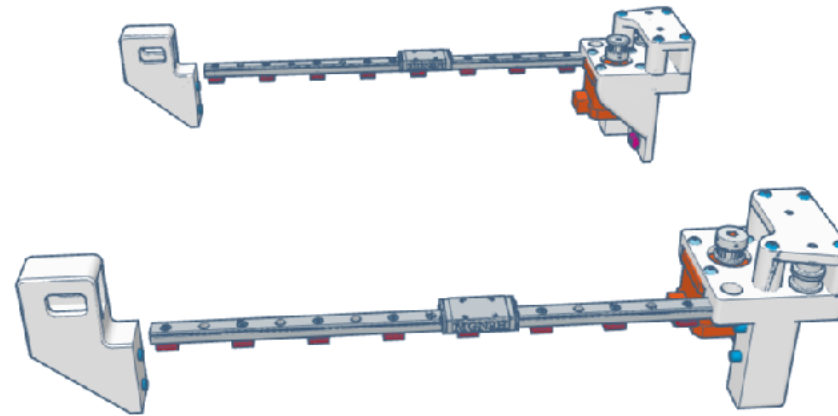


(4 M4x35 FHCS and 4 M4 hex nut total)

Building the AB kinematics

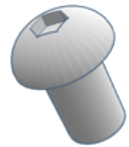
AB gantry

22



M3x30 BHCS screw

8



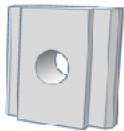
M3x8 BHCS screw

24



M3x6 BHCS screw

4



M3 T-slot nut

16



M3x30 pin

4

M3x20 pin

2



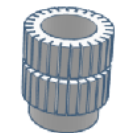
M3x0.5 spacer

26



M3x10 spacer

2



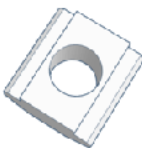
M3 Heat set insert

12



M5x8 BHCS screw

12



M5 T-slot nut

2



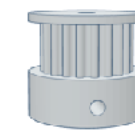
MGN9H 300mm rail
or MGN12H 300mm rail

2



GT2 16T6W3B toothless idler

8



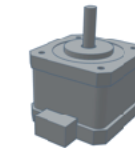
GT2 16T6W5B motor pulley

2



M5 Nylstop nut
(from Ender 3)

2

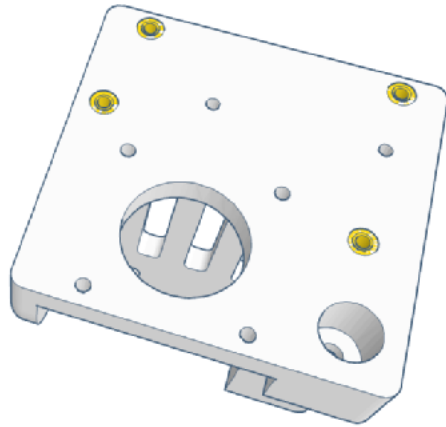


Nema 17 42-40
(from Ender 3)

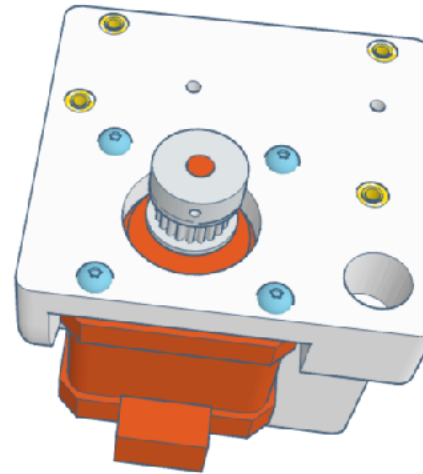
2

Building the AB kinematics

23



Prepare the bottom part of AB motor holders:
4 heat-set inserts each side
(8 M3 heat-set inserts total)

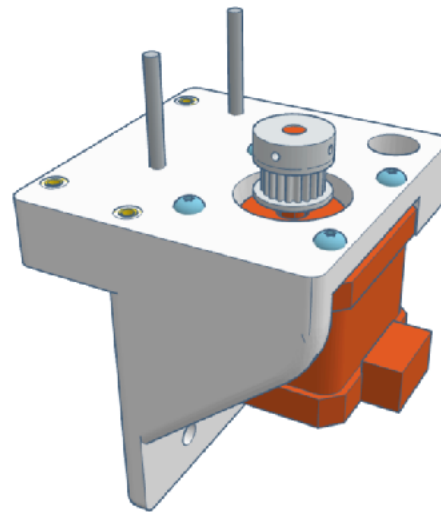


Install the motors under the plates with 4 M3x8 BHCS screws in the 4 corners

Mind the motor plug orientation
(8 M3x8 BHCS total)

Also place the motor pulleys onto the shafts of A & B motors

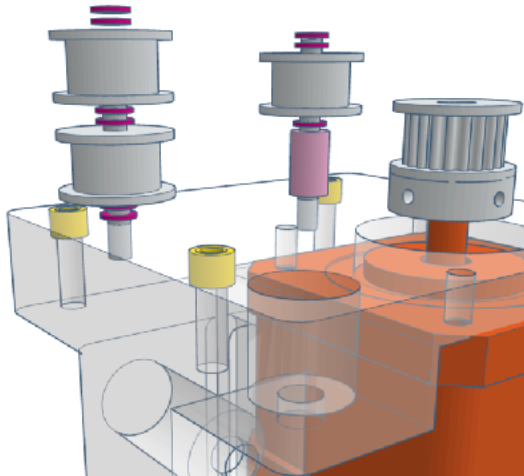
Do not tighten their set screws too much. They will need to be adjusted after belts installation



Carefully press-fit two M3x30mm pins into their holes, making sure to keep them straight up
(4 M3x30mm pins total)

Building the AB kinematics

24

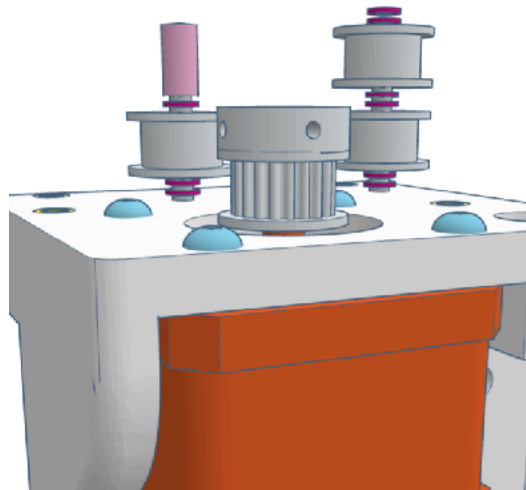


For the Left AB mount:

Stack
2 M3x0.5mm spacers,
1 GT2 16T 6mm toothless idler,
2 M3x0.5mm spacers,
1 GT2 16T 6mm toothless idler,
2 M3x0.5mm spacers onto the outer most pin.

Stack
1 M3x10mm spacer,
1 M3x0.5mm spacer,
1 GT2 16T 6mm toothless idler,
2 more M3x0.5mm spacers on the inner-most pin.

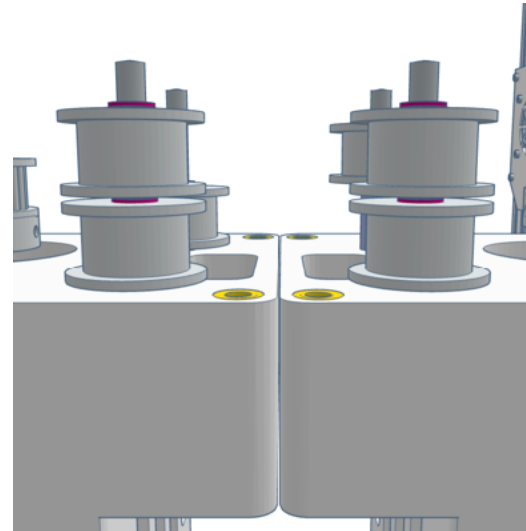
The best option is certainly to buy and install a metal M3x10 spacer, but should you not have one on hand, you may use the printed one provided in the stl files.



For the Right AB mount:

Stack
2 M3x0.5mm spacers,
1 GT2 16T 6mm toothless idler,
2 M3x0.5mm spacers,
1 GT2 16T 6mm toothless idler,
2 M3x0.5mm spacers onto the outer most pin.

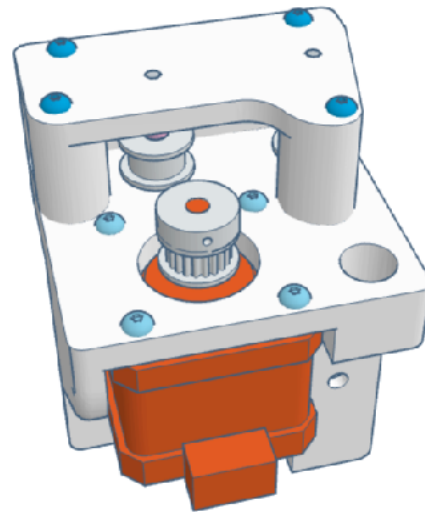
Stack
2 M3x0.5mm spacers,
1 GT2 16T 6mm toothless idler,
1 M3x0.5mm spacer,
1 M3x10mm spacer on the inner-most pin.



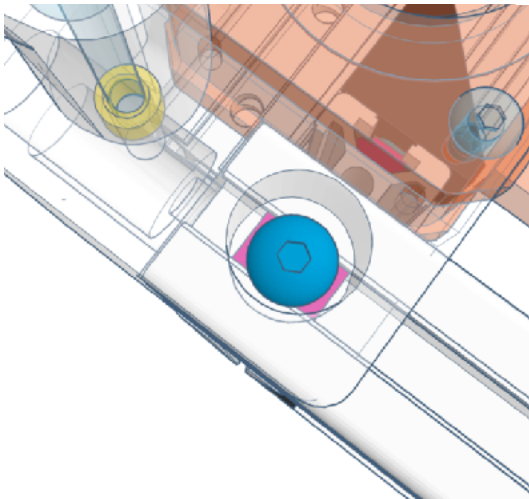
When the motors are one against another, the idlers should be perfectly level between both sides. If that is not the case, adjust the shims count as necessary

Building the AB kinematics

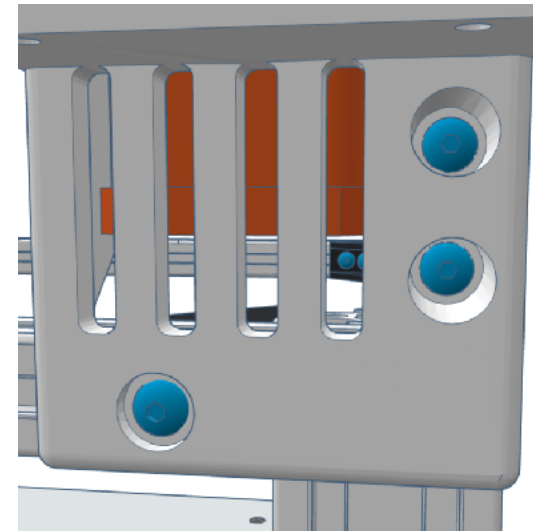
25



Place the top half of the AB motors mount on each side, carefully inserting the M3x30 pins top into their slots.
Attach them with 4 M3x30 BHCS for each side (8 M3x30 BHCS total)



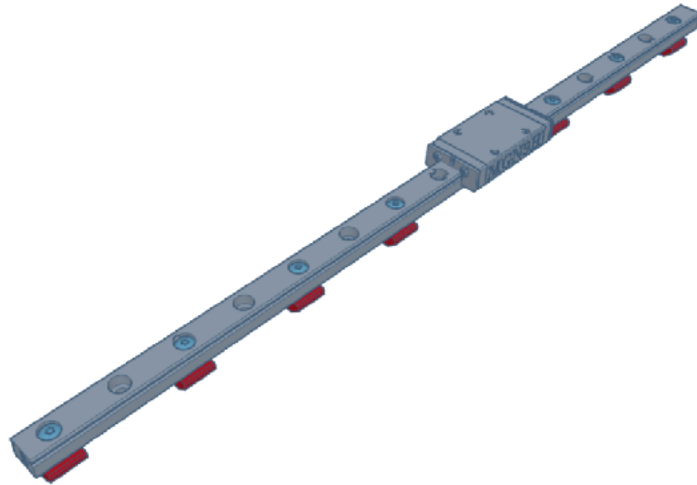
Slide a M5 T-slot nut in each Y axis and install the AB motor holders using a M5x8 BHCS screw
(2 M5 T-slot nuts and 2 M5x8 BHCS total)



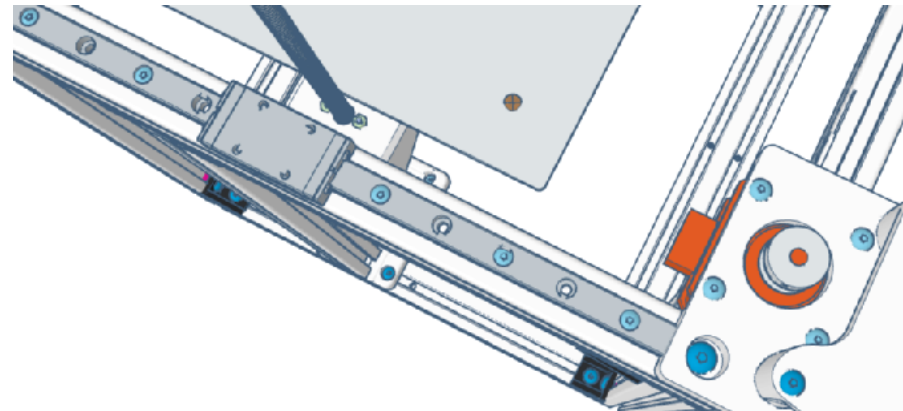
Use 2 M5x8 BHCS screws to hold into the tapped end of the 2040-C Y axis and another one to hold onto the back brace M5 T-slot nut
(6 M5x8 BHCS total)

Building the AB kinematics

26



Install 8 M3 T-slot nuts, one every other hole, securing them with 8 M3x8 BHCS screws through the rail. Leave them quite loose for now. (16 M3 T-slot nut and 16 M3x8 BHCS total)

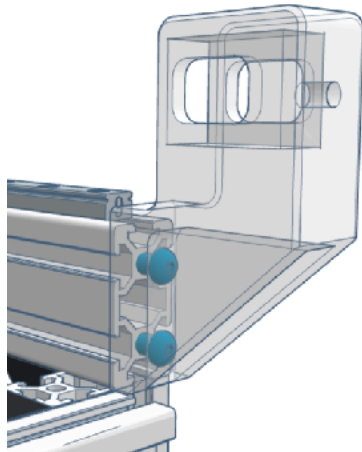


Slide the T-slot nuts into the top of each Y extrusion, aligning them one after another, all the way to the AB motor holders. Use a centring 3D printed jig to center them and slightly tighten the screws, just enough to hold the rail in place.

DO NOT FULLY TIGHTEN THE SCREWS YET

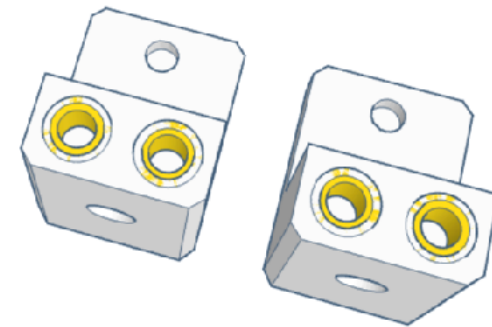
Building the AB kinematics

27

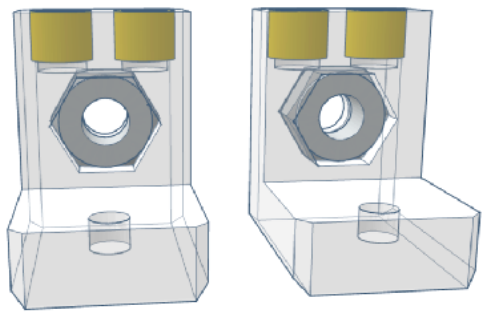


Install the front holders with 2 M5x8 BHCS screws each side
(4 M5x8 BHCS total)

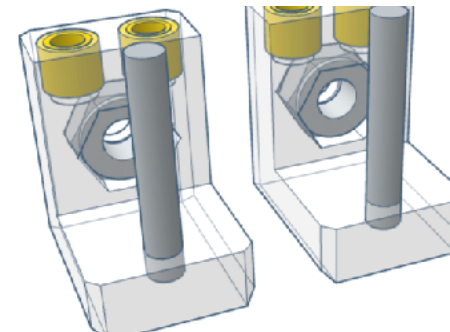
Mind the orientation, the holders are not symmetrical



Prepare both the Bottom_Left and Bottom_Right belt tensioners by placing 2 M3 heat-set inserts in the top holes
(4 M3 heat-set inserts total).



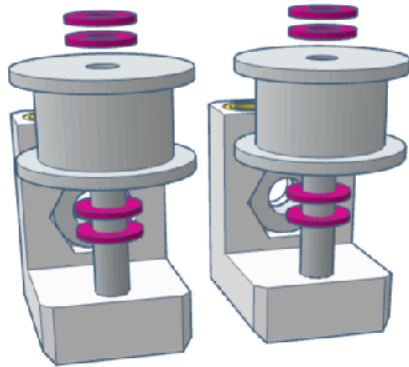
Insert an M5 Nylstop nut into the groove inside both holders. A small dab of glue may help keep them in place
(2 M5 hex nut total).



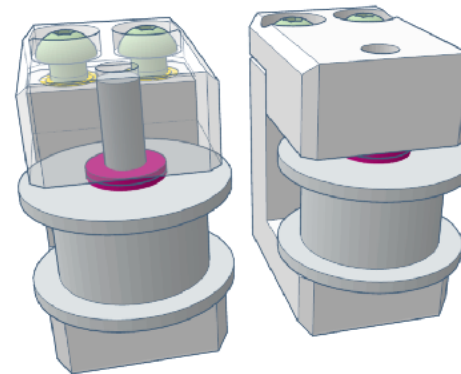
Press-fit the M3x20mm pins into each side slot, making sure they go square
(2 M3x20mm pins total)

Building the AB kinematics

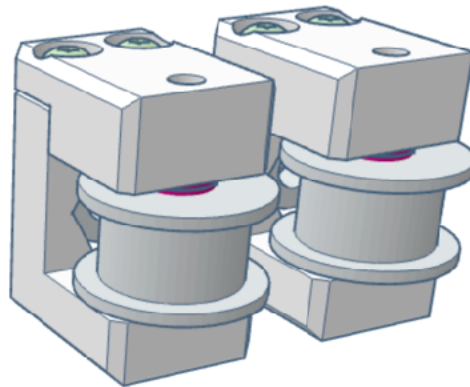
28



Stack 2 M3x0.5mm shims, a 16mm toothless idler and 2 M3x0.5mm shims onto each pin (8 M3x0.5 shims and 2 GT2 16T6W3B toothless idlers).



Carefully place the Top_Left and Top_Right parts onto their pins and use 2 M3x6 BHCS screws each side to secure the assembly (4 M3x6 BHCS total)

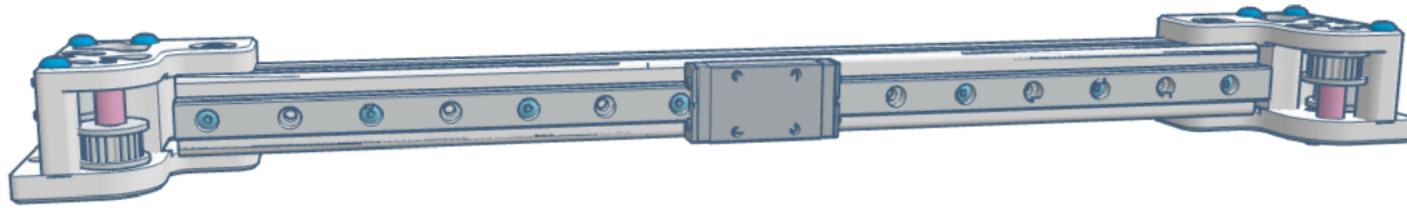


Set the two tensioners aside, it will be way easier to install them with the belts.

Building the X axis

X axis gantry

29



M3x30 BHCS screw

6



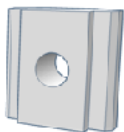
M3x8 BHCS screw

8



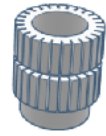
M3x6 BHCS screw

8



M3 T-slot nut

8



M3 Heat set insert

6



M3x10 spacer

8



M3x0.5 spacer

8



M3x30 pin

8



MGN9H 300mm rail
MGN12H 300mm rail

1

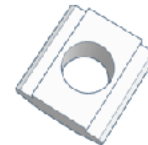
6



M5x8 BHCS screw

6

4



M5 T-slot nut

6

12



GT2 16T6W3B toothless idler

2

4

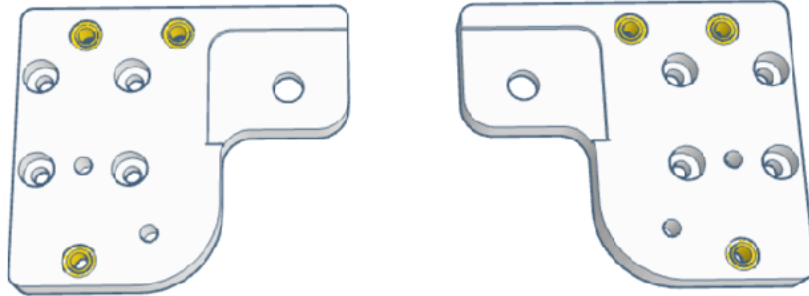


GT2 16T6W3B toothed idler

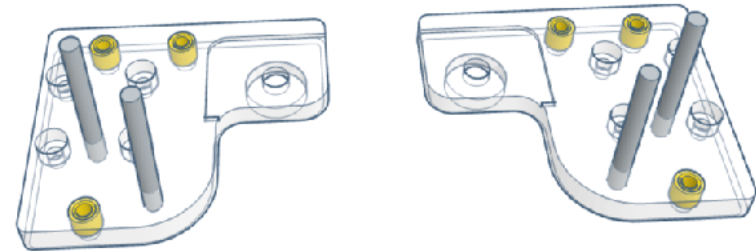
2

Building the X axis

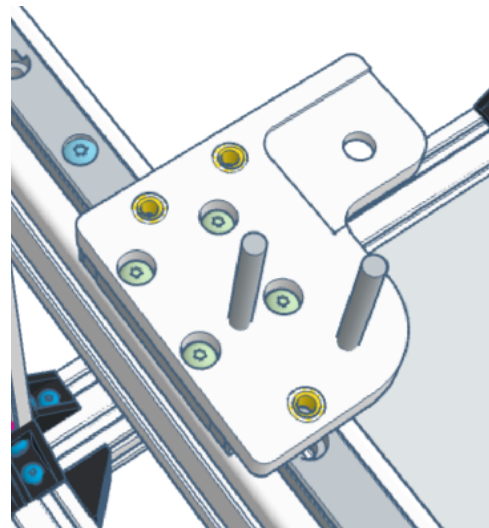
30



Prepare both bottom parts of the X carriage with 3 heat-set inserts each side (6 M3 heat-set inserts total).



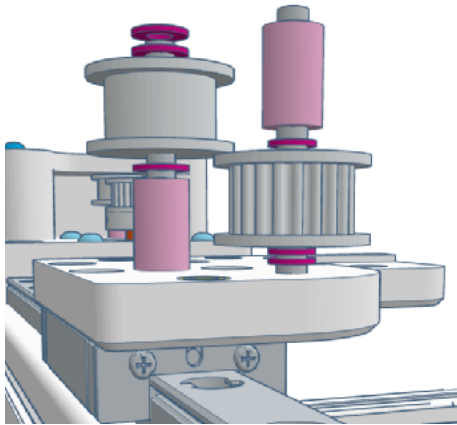
Carefully press-fit the M3x30 pins into their slots, keeping them square to the base (4 M3x30mm pins total)



Place each part onto its rail carriage and attach them with 4 M3x6 BHCS screws each side (8 M3x6 BHCS total)

Building the X axis

31



For the left hand side:

Left-most pin:

- 1 M3x10mm spacer,
- 1 M3x0.5mm shim,
- 1 **toothless** idler,
- 2 M3x0.5mm shims

Right-most pin:

- 2 M3x0.5mm shims,
- 1 **toothed** idler,
- 1 M3x0.5mm shim,
- 1 M3x10mm spacer

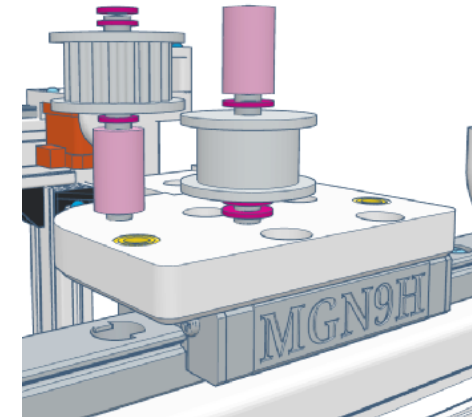
For the right hand side:

Left-most pin:

- 1 M3x10mm spacer,
- 1 M3x0.5mm shim,
- 1 **toothed** idler,
- 2 M3x0.5mm shims

Right-most pin:

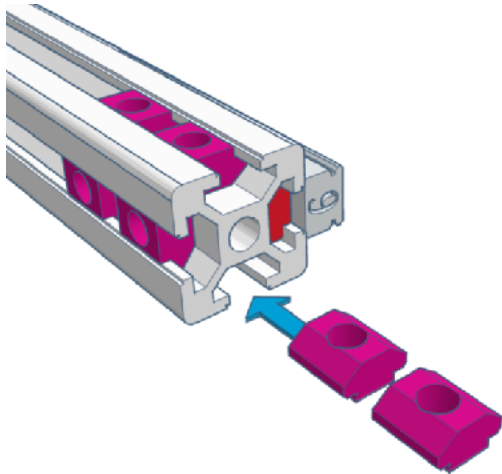
- 2 M3x0.5mm shims,
- 1 **toothless** idler,
- 1 M3x0.5mm shim,
- 1 M3x10mm spacer



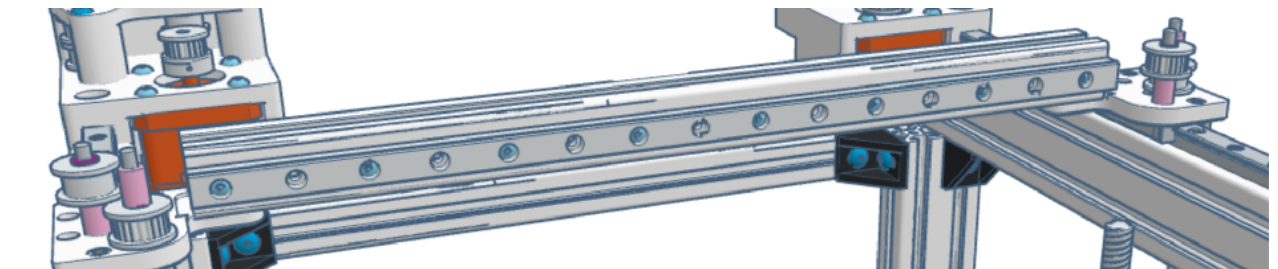
Insert 8 M3 T-slot nuts into the front of the 300mm 2020 extrusion, the last rail, and secure the assembly with 8 M3x8 BHCS screws

Building the X axis

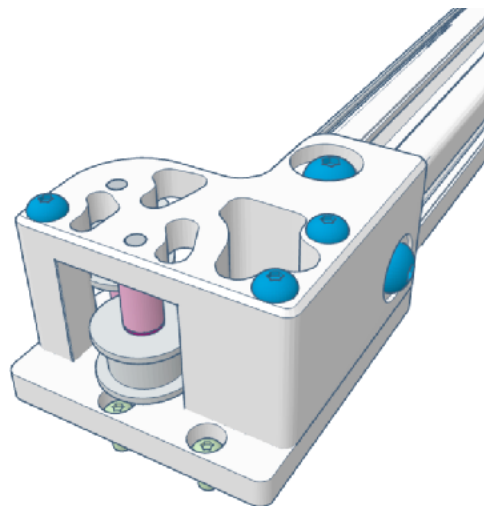
32



Insert 2 M5 T-slot nuts on the top side, 2 on the back side and 2 more under the extrusion



Place the X axis onto both carriages



Carefully push the top halves onto the M3x30 pins.

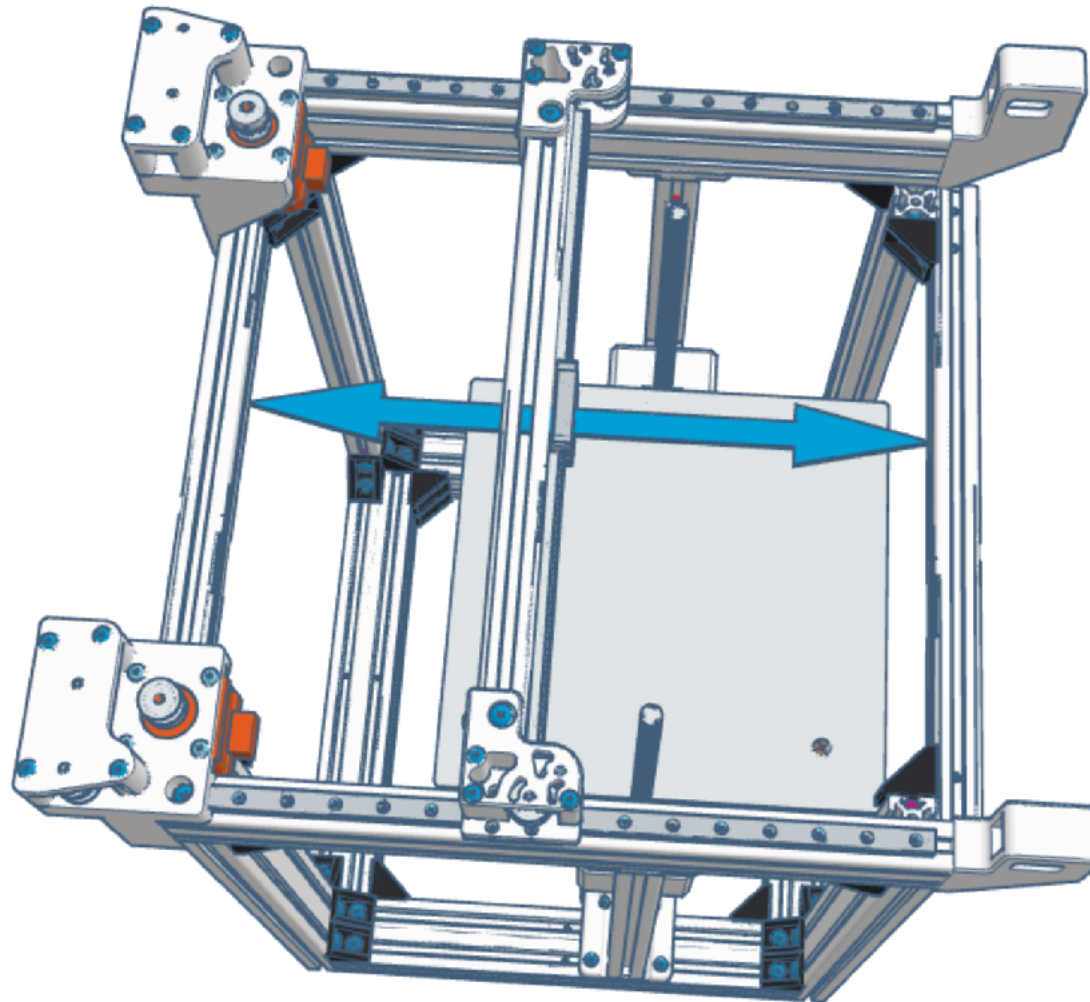
Secure each carriage with 3 M3x30 BHCS screws

Center and secure the X extrusion with 3 M5x8 BHCS on each end: one on the top, one on the back and one under the extrusion.

(6 M3x30 BHCS and 6 M5x8 BHCS total)

Building the X axis

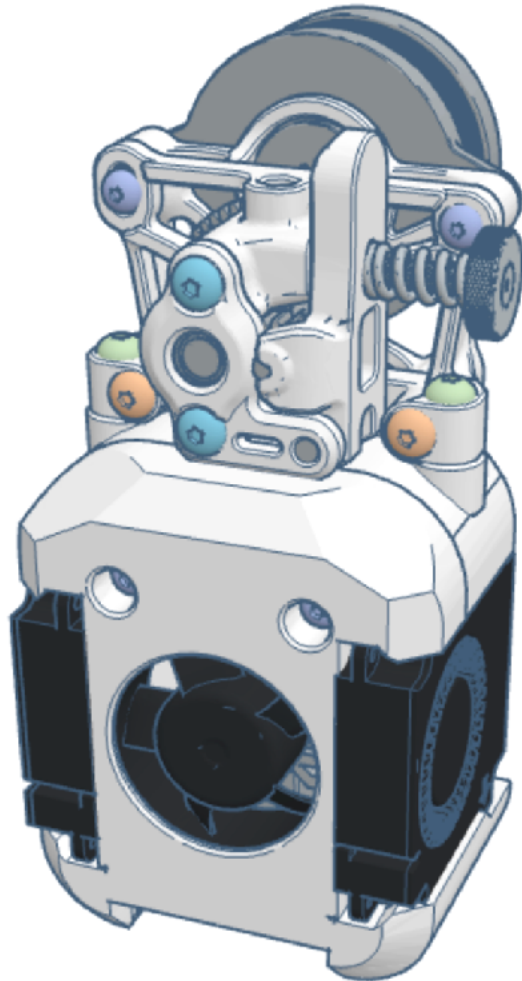
33



Finally, slide the X axis back and forth several times so the Y rails align themselves
Gradually tighten the M3x8 screws each side from one end to another to lock the rails in place

Toolhead assembly

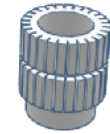
Dragonburner toolhead



Why this toolhead?

All this hardware will be reused for MedusaHC.
If you are building for Medusa, please refer to the relevant manual for your toolhead.

Dragonburner



34
M3 Heat set insert

12



M3 Hex nut

1



3010 axial fan

1



4010 radial fan

2



Bambulab X1 series hotend

1



M3x20 BHCS screw

4



M3x12 BHCS screw

3



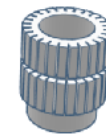
M3x8 BHCS screw

4



M3x6 BHCS screw

4



M3 Heat set insert

5



M3x16 BHCS screw

2



M3x14 BHCS screw

2



M3x12 BHCS screw

2



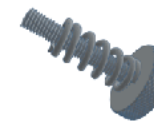
M3x8 BHCS screw

2



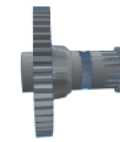
Nema 14 motor 10t

1



Thumbscrew

1



Filament gear assembly

1



Idler gear assembly

1



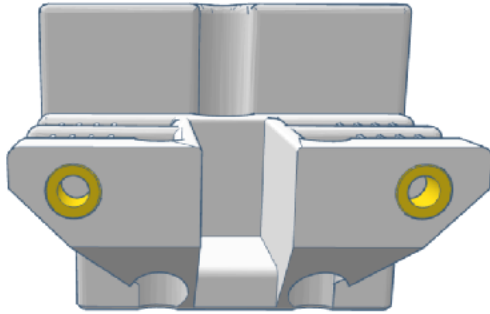
M3x20 pin

1

Sherpa mini

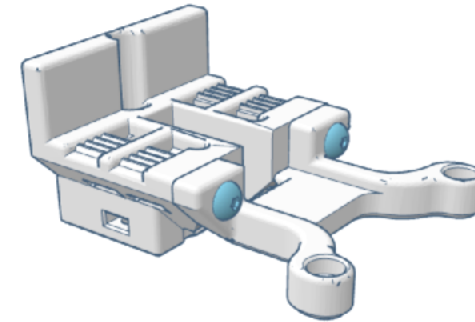
Toolhead assembly

35

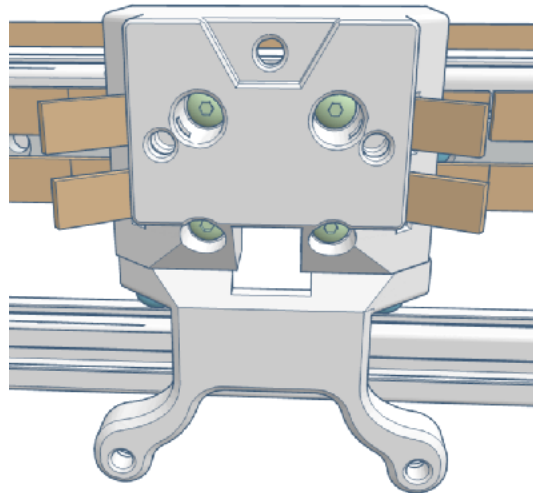


Prepare the top part of the carriage with 2 M3 heat-set inserts.

Remember to break the top front support piece.



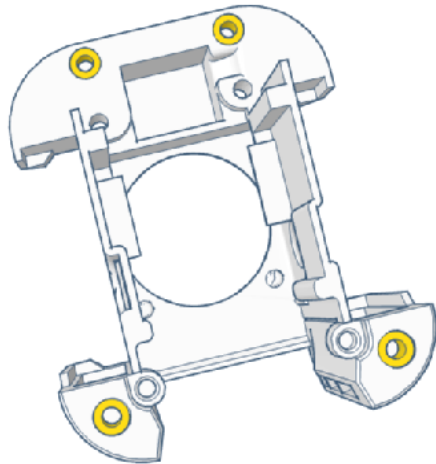
Use 2 M3x8 BHCS to attach the top and bottom parts together.



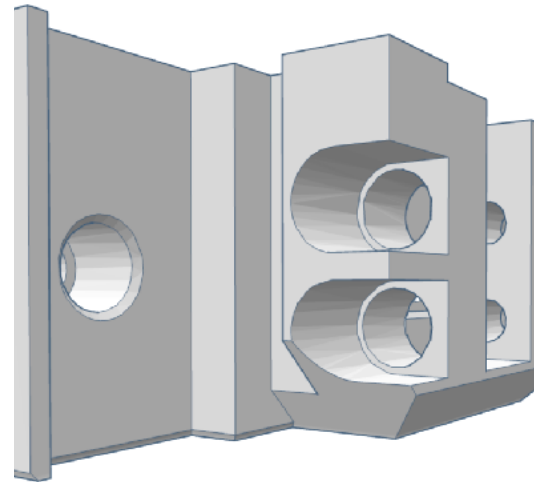
Set the carriage aside, it will be used when installing the belts (use 4 M3x6 BHCS to attach it to the X rail).

Toolhead assembly

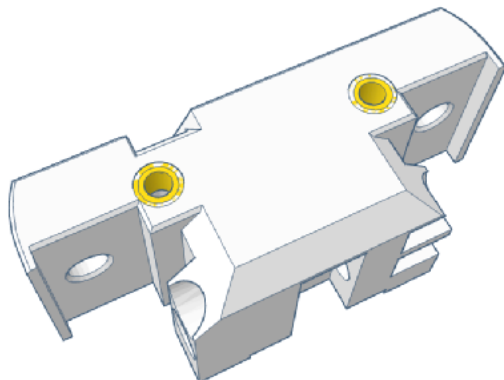
36



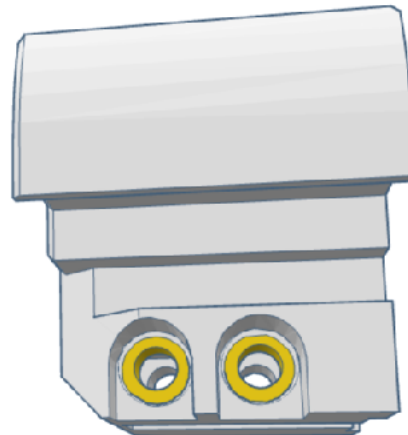
Place 4 M3 heat-set inserts in the back of the toolhead cowling.



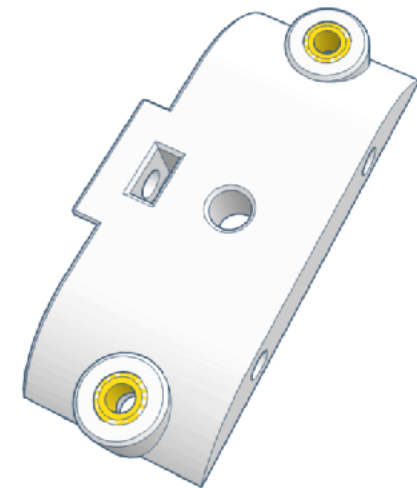
Break the supports on the side of the hotend support



Install 2 M3 heat-set inserts into the top holes of the hotend support.



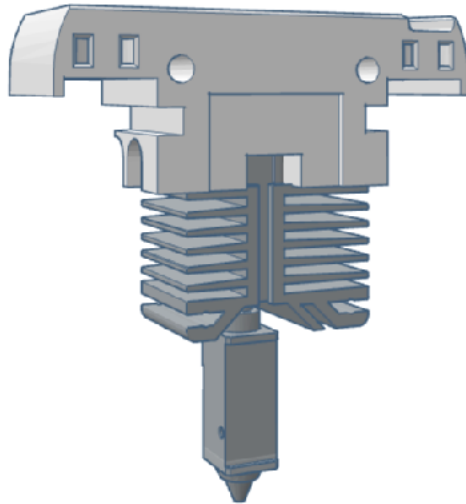
Add 2 M3 heat-set inserts to the side, in the holes that were covered by the supports.



Install 2 M3 heat-set inserts into the top holes of the extruder adapter plate.

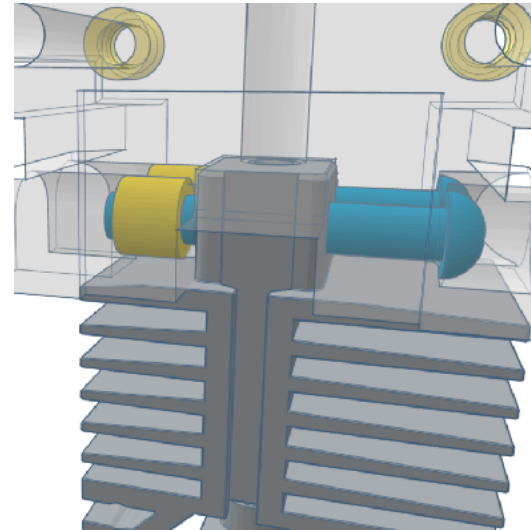
Toolhead assembly

37

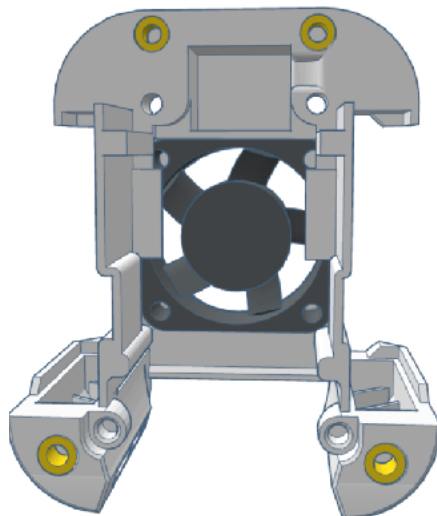


Install the Bambulab X1 series hotend with its heatsink under the hotend support.

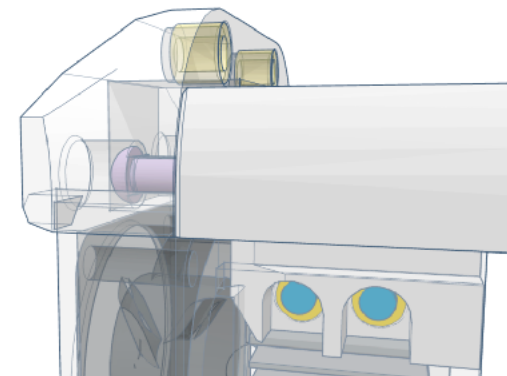
Mind the orientation of the heatsink



Use 2 M3x20 BHCS to lock the hotend in place



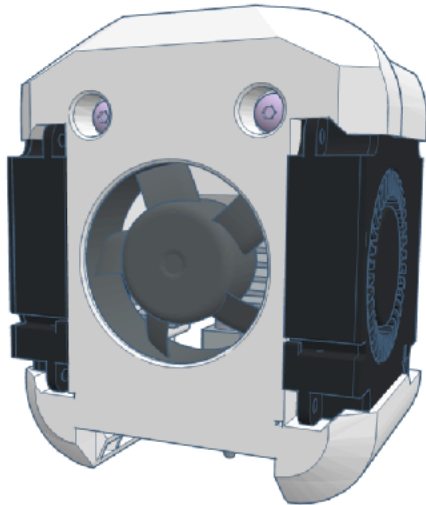
Press-fit the 3010 axial fan into the front of the cowling
Remember to route the wires towards the back.



Slide the hotend support with the hotend into the back of the cowling, then use 2 M3x12 BHCS from the front to secure it.

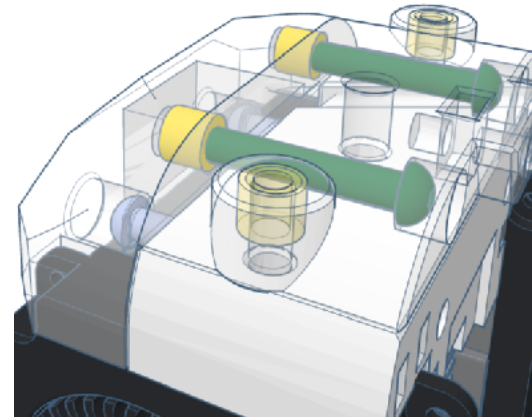
Toolhead assembly

38

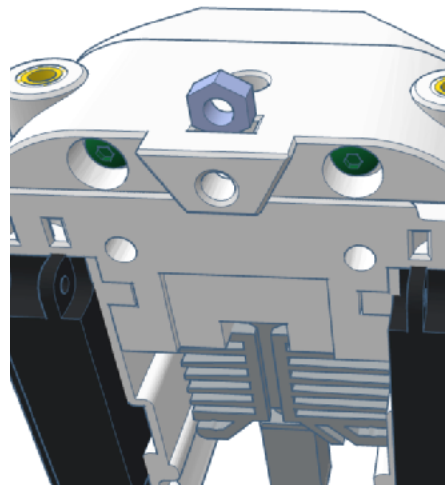


Press-fit the 2 4010 radial fans in each side of the cowling, from back to front.

Keep the wiring out towards the back.



Place the extruder support on top of the hotend, behind the cowling, and pass 2 M3x20 BHCS screws to hold it.

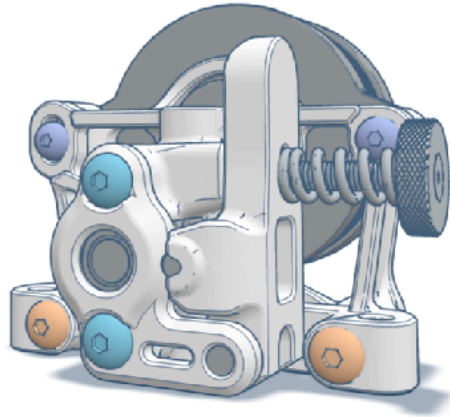


Drop 1 M3 hex nut into its slot.

You may want to hold it with a drop of glue, in that case, temporarily add a screw from the back to hold the nut in place while the glue cures.

Toolhead assembly

39

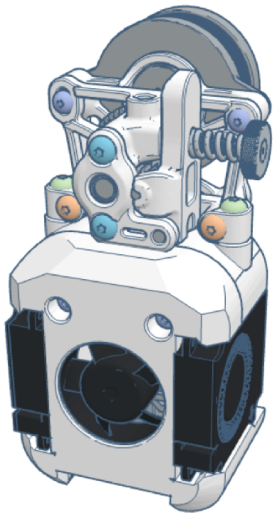


Now is the time to build your Sherpa mini extruder.

Its author provides an excellent assembly manual.

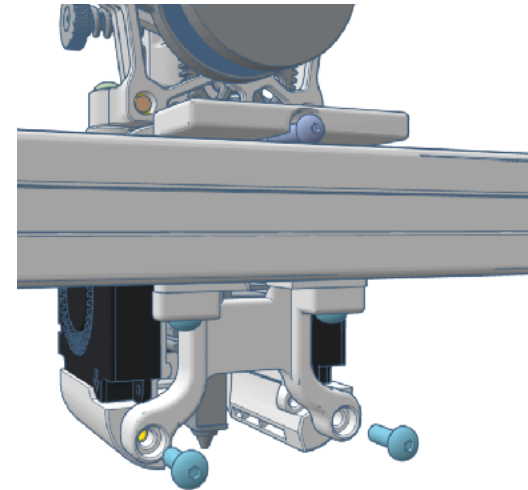
Download it from the GitHub linked below, proceed with the build, then come back for the last assembly steps.

[https://github.com/Annex-Engineering/Sherpa_Mini-Extruder/tree/master/Build Instructions](https://github.com/Annex-Engineering/Sherpa_Mini-Extruder/tree/master/Build%20Instructions)



Place your newly assembled extruder on top of the Dragonburner.

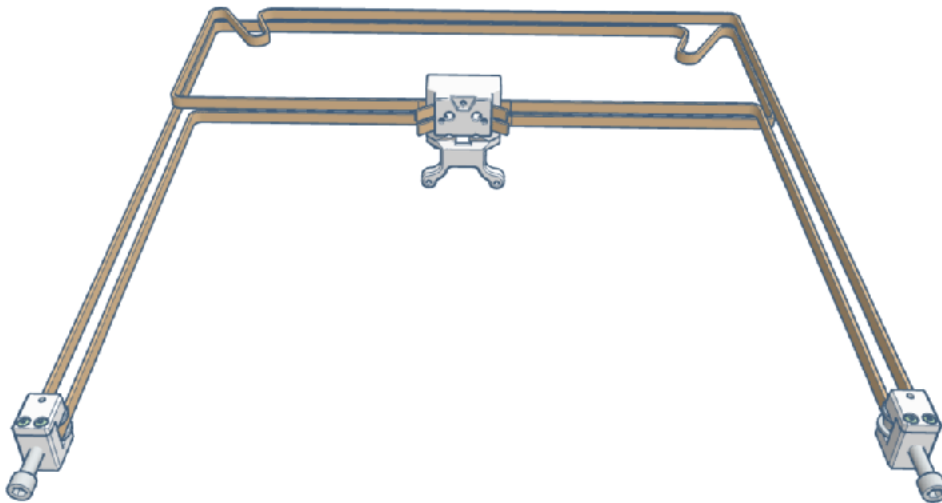
Use 2 M3x14 BHCS to attach the Sherpa mini to the extruder plate.



After completing the next section and installing the belts, your toolhead will attach to its carriage from the back, with 2 M3x8 BHCS on the bottom, and 1 M3x12 BHCS at the top.

A & B belts installation

40



GT2 6mm belt approx. 2x1,5m



M5x25 SHCS
(from Ender 3)

2

"Equal belt tension is important to the proper function of a CoreXY motion system."

We recommend to run one belt to get the required length, remove the belt from the printer and cut the second belt to the exact same length.

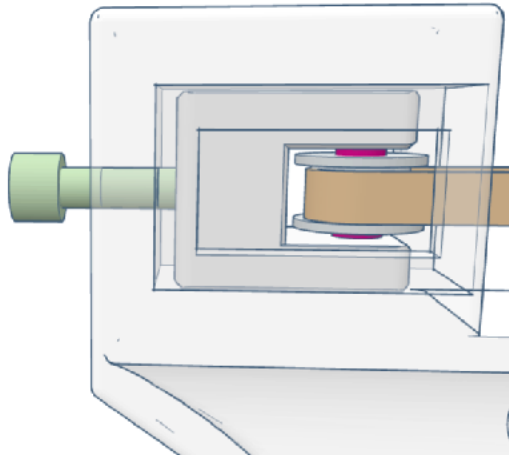
As both belt paths have the same length this is an easy way of getting a consistent tension."

- The Voron team

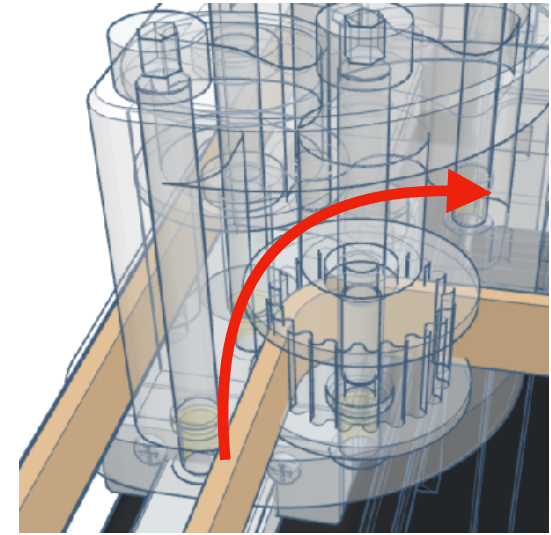
A & B belts installation

41

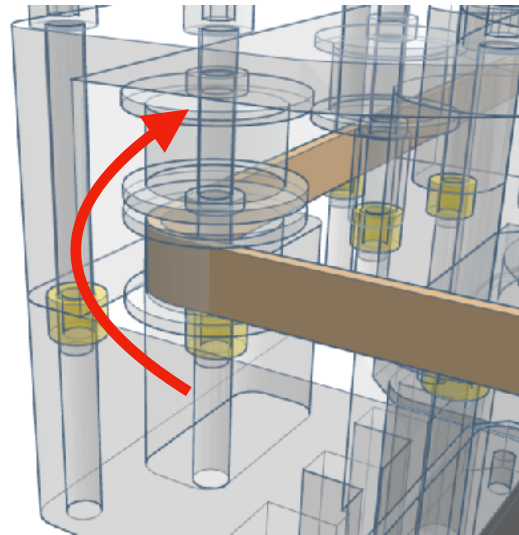
A belt



Pass the belt into the L side tensioner (flat side against idler) then insert the tensioner into the L front holder, and secure it with a M5x25 SHCS screw passing from the front of the holder



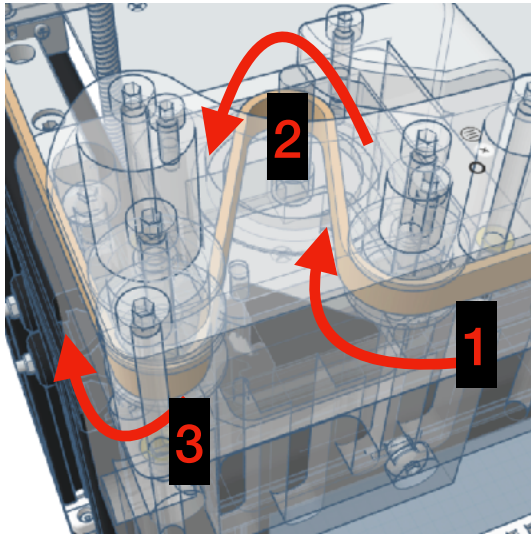
Pass the short side of the belt into the toothed idler and towards the X axis toolhead carriage



The longer side passes next to the left Y carriage all the way to the AB motor holder at the back left
Use the lower idler to turn 90° and head to the right AB motor holder

A & B belts installation

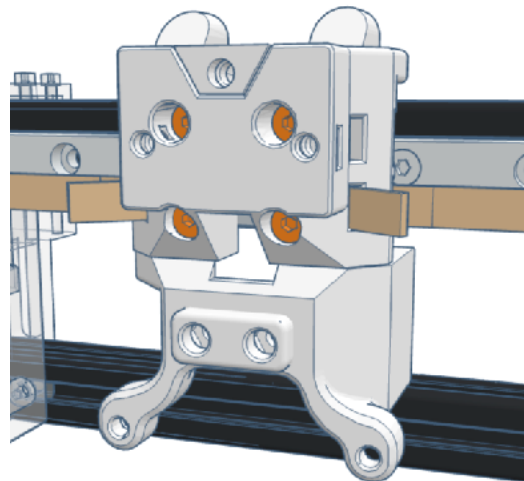
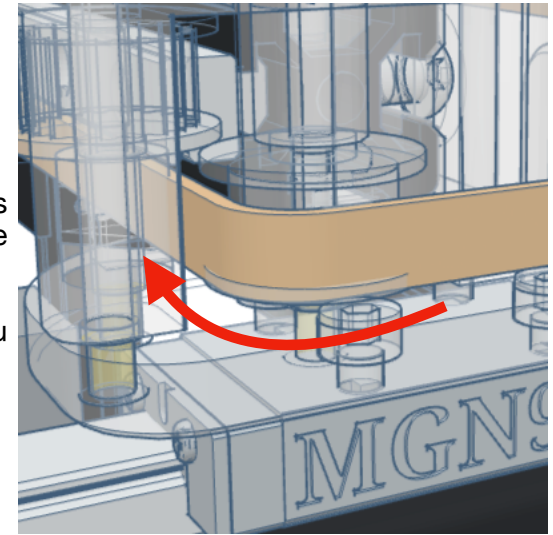
42



- 1:** Into the right AB motor holder, use the first idler to redirect the belt towards the motor pulley (not shown here)
- 2:** Place the belt into the A motor pulley, **it's time to adjust pulley height.** Then back to the second idler (the lower one) to
- 3:** continue towards the front of the printer

Finally, use the lower idler inside the right Y axis carriage to turn 90° and head to the right side of the printhead carriage

You will need to use whichever toolhead carriage you have chosen to hold both extremities of the belt



Use the toolhead carriage to temporarily clamp the belt. Use 4 M3x6 BHCS to install it onto the rail carriage.

Once secured, cut the belt to the correct length, and unmount it to use as reference

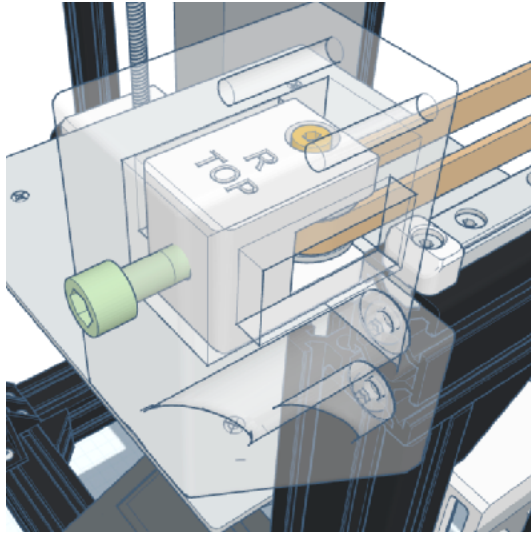
Place the freshly cut A belt on top of the remaining GT2 length and make sure they stay perfectly flush (you can tape them together)
Cut the second belt at the exact dimension: matching teeth count now ensures matching tension later

You can now reinstall the A belt with the same exact procedure you just followed

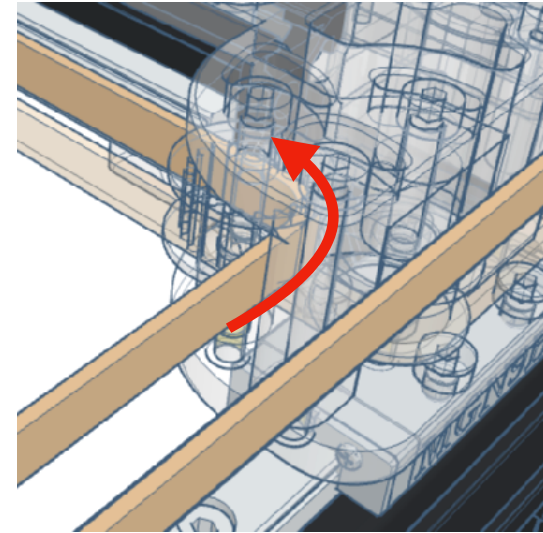
A & B belts installation

B belt

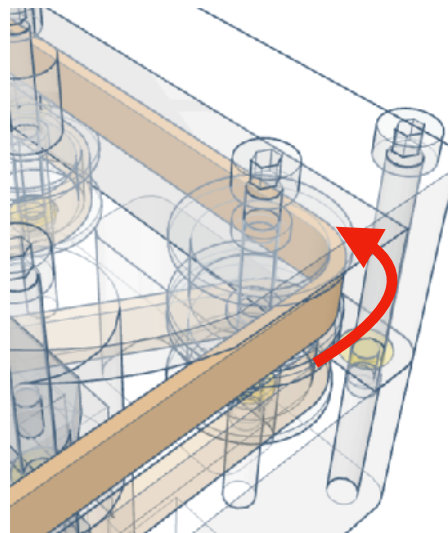
43



Pass the belt into the R side tensioner (flat side against idler) then insert the tensioner into the R front holder, and secure it with a M5x25 SHCS screw passing from the front hole of the holder



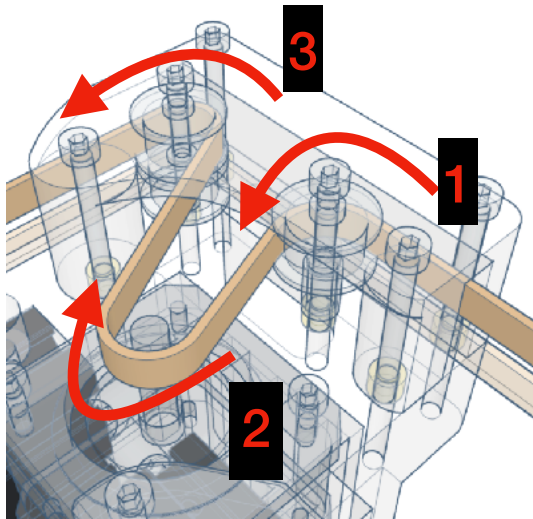
Pass the short side of the belt into the toothed idler and towards the X axis toolhead carriage



The longer side passes next to the right Y carriage all the way to the AB motor holder at the back right
Use the upper idler to turn 90° and head to the left AB motor holder

A & B belts installation

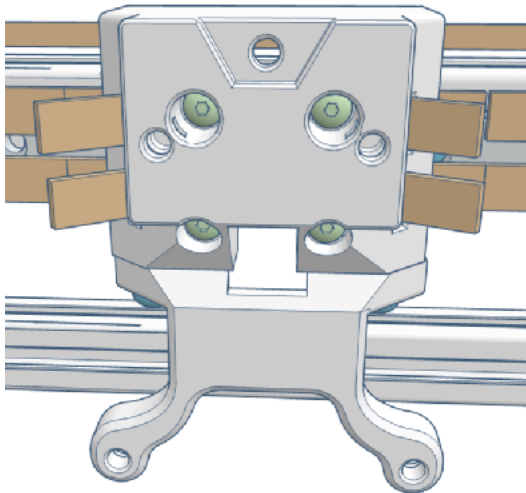
44



1: Into the left AB motor holder, use the first idler to redirect the belt towards the motor pulley (not shown here)

2: Place the belt into the B motor pulley, **it's time to adjust pulley height**. Then back to the second idler (the upper one)

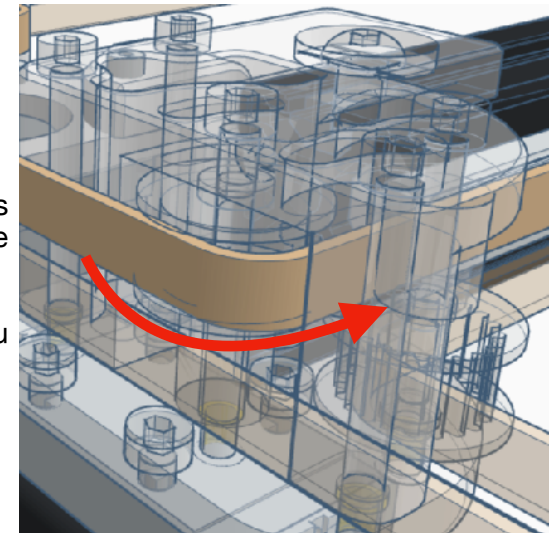
3: continue towards the front of the printer



Use the X carriage to clamp the belts, by loosening slightly, tensioning the belts, and tightening it back.

Finally, use the upper idler inside the left Y axis carriage to turn 90° and head to the left side of the printhead carriage

You will need to use whichever toolhead carriage you have chosen to hold both extremities of the belt



IMPORTANT: Double check that both belts are perfectly level and straight all along their path

If that is not the case, you may need to adjust the number of M3x0.5mm spacers used to level the idlers

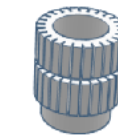
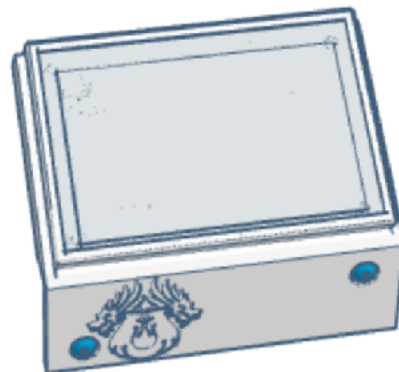
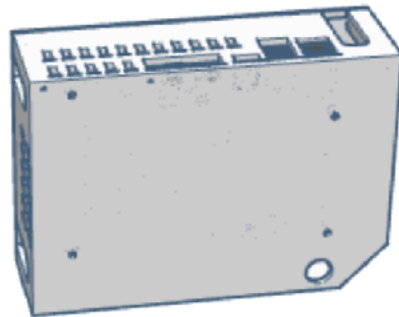
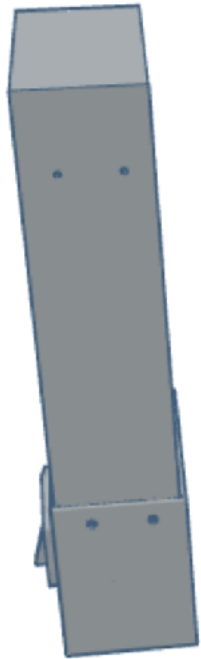
Also check for the A and B motors pulleys height. You may have to unscrew them a bit to align them with the belts

It may be a good idea to add a drop of blue threadlocker to the set screws of the pulleys before final tightening

PSU, control box & screen

45

Power supply, control box and screen housing



M3 Heat set insert

8



M5x8 BHCS screw

8



BTT HDMI5 V1.2

1



24V 15A PSU
(from Ender 3)

1

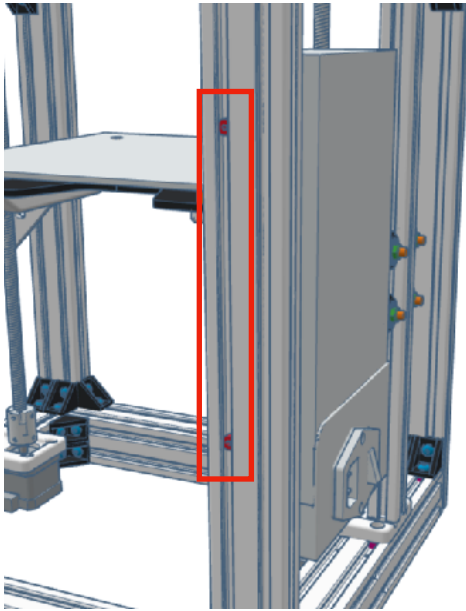


M4x20 SHCS
(from Ender 3)

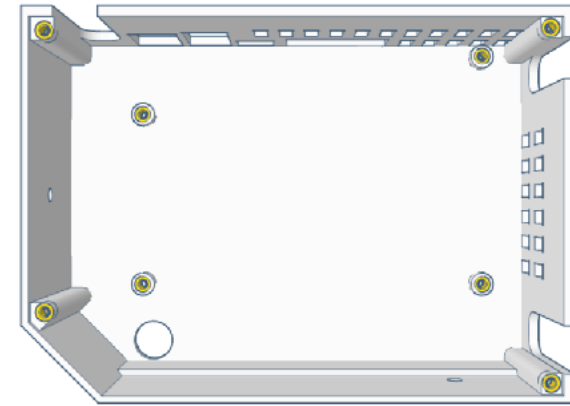
2

PSU, control box & screen

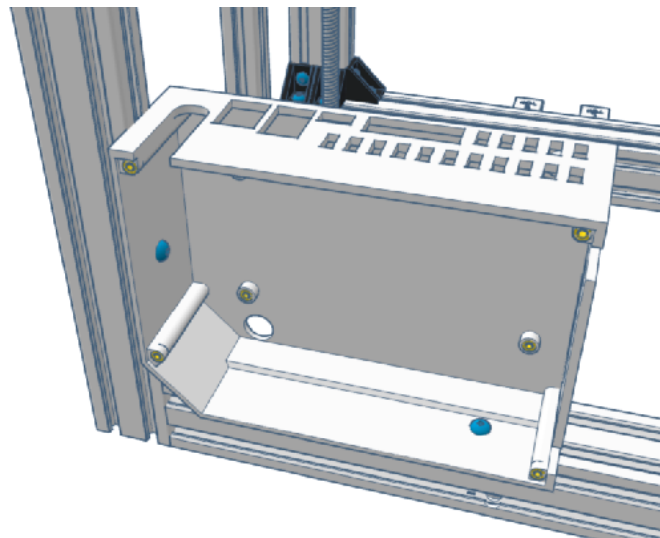
46



Use original M4x20 SHCS screws to attach the original power supply to the back left 2040-B Z post



Prepare the bottom of the electronics box by inserting 8 heat-set inserts, one into each corner hole, and on the inside mainboard pillars

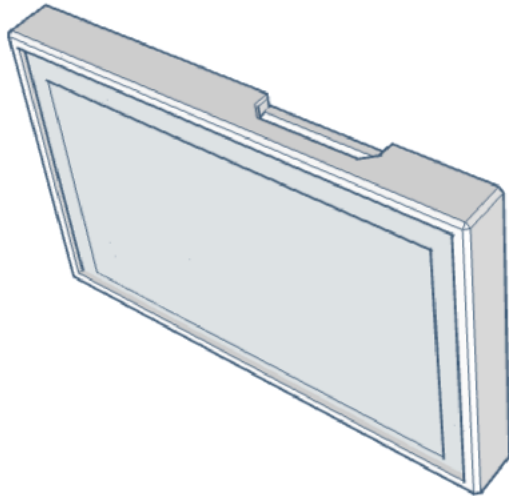


Place the electronics box in the back lower right corner of the frame, and use the two remaining M5 T-slot nuts to secure it with 2 M5x8 BHCS screws

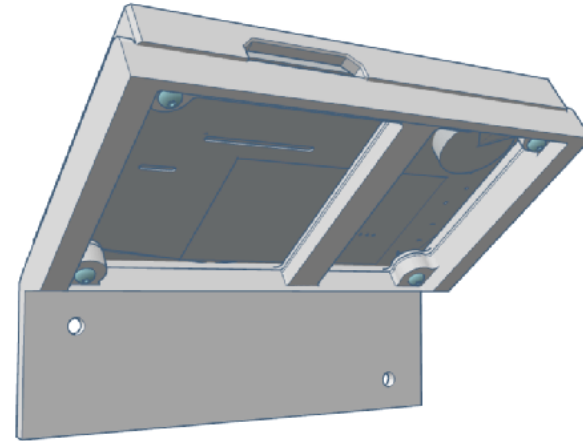
Box shown is for BTT Manta 5p mainboard. You may need a different model depending on your configuration

PSU, control box & screen

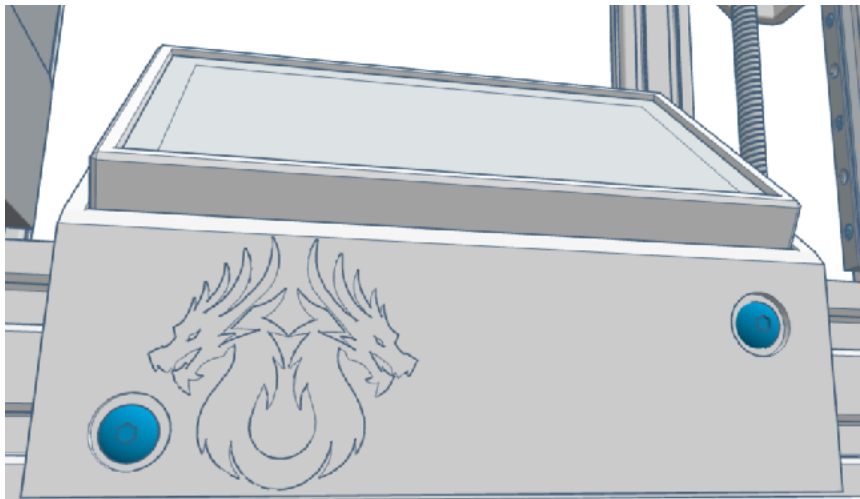
47



Start by inserting your BTT HDMI5 screen into the frame.
The rear speaker and the buttons indicate the correct orientation.



Place the assembly onto the main support part,
and attach the screen from the underside with 4
M3X8 BHCS screws.



Mount the completed screen housing to the front
of your Duender using the last two remaining M5
T-slot nuts with 2 M5x8 BHCS screws

Mainboard installation

Wiring

48

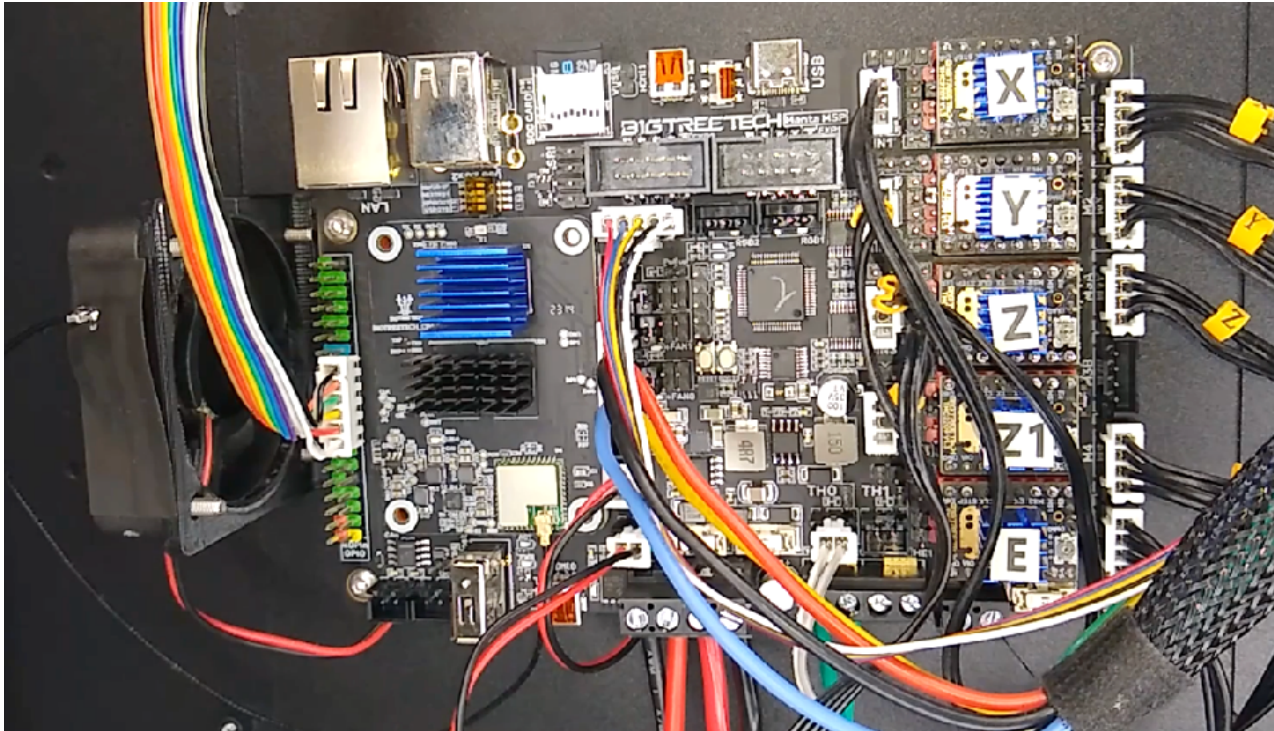


Image from detailed videos about the Manta 5P by Victor Bared

Jumpers setup, image install: <https://2cm.es/1IYBp>

Wiring: <https://2cm.es/1IYBr>

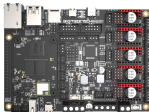
This section is intended for the BTT Manta 5P mainboard loaded with a CB1/CB2

If you are building Medusa right away, please check M8P wiring and setup in the MHC manual.

You will find all the information you need on the BTT Wiki: <https://2cm.es/1IYBC>

Files and firmware can be downloaded from the GitHub: <https://2cm.es/1h2DL>

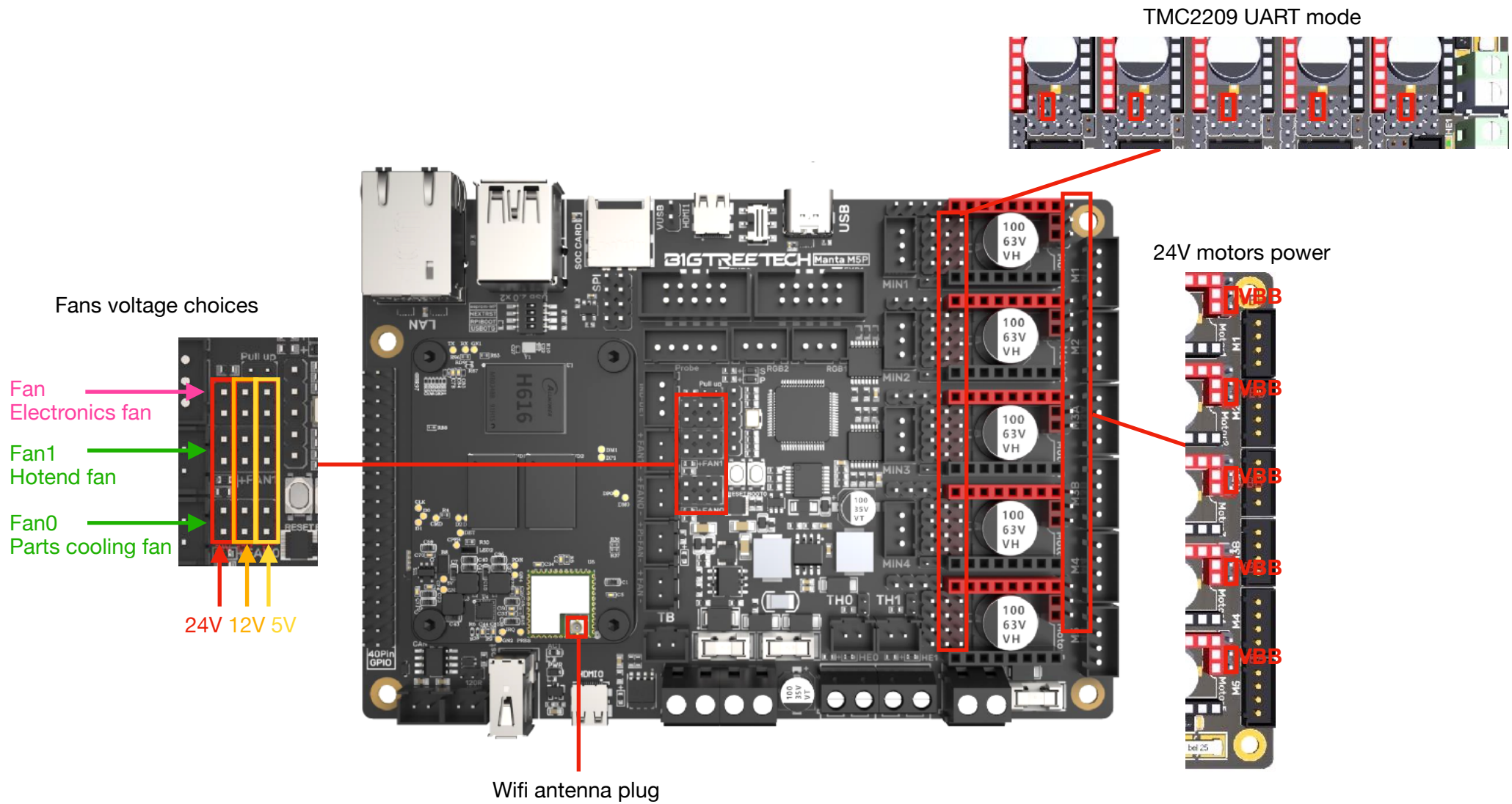
Keep in mind you may need to extend or completely remake some cables, and for this some tools like a soldering iron, good crimping pliers and a set of connectors will prove more than useful

	M3x6 BHCS screw	8
	M3x18 BHCS screw	8
	M3 Nylstop nut	8
	BTT Manta 5P BTT Manta 8P V2.0	1
	BTT CB1 OR CB2	1
	TMC2209 driver	5
	4010 axial fan (from Ender 3)	2
	Wires in different colours 1 mm ² , 0.5 mm ² etc.	plenty
	Micro-JST connectors and pins	some

Mainboard installation

Jumpers basics

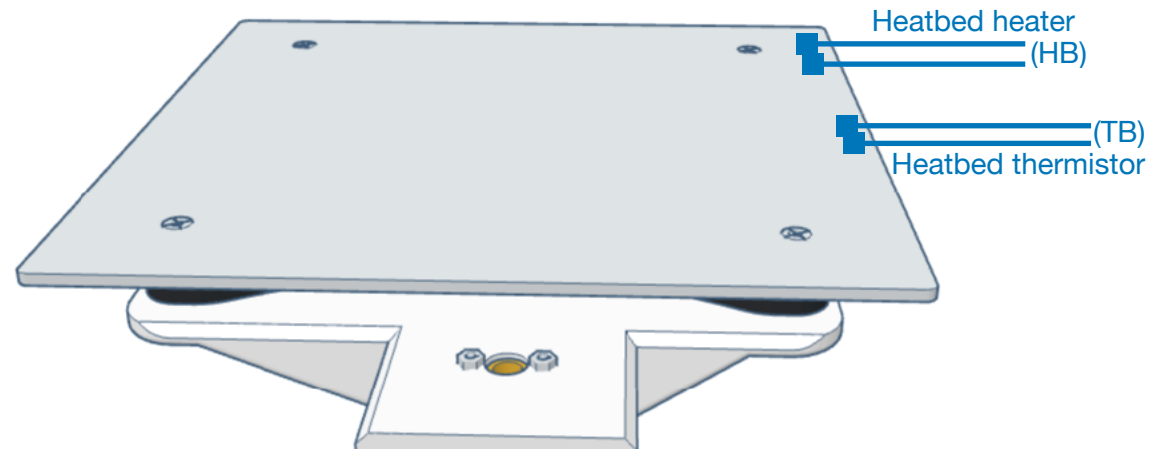
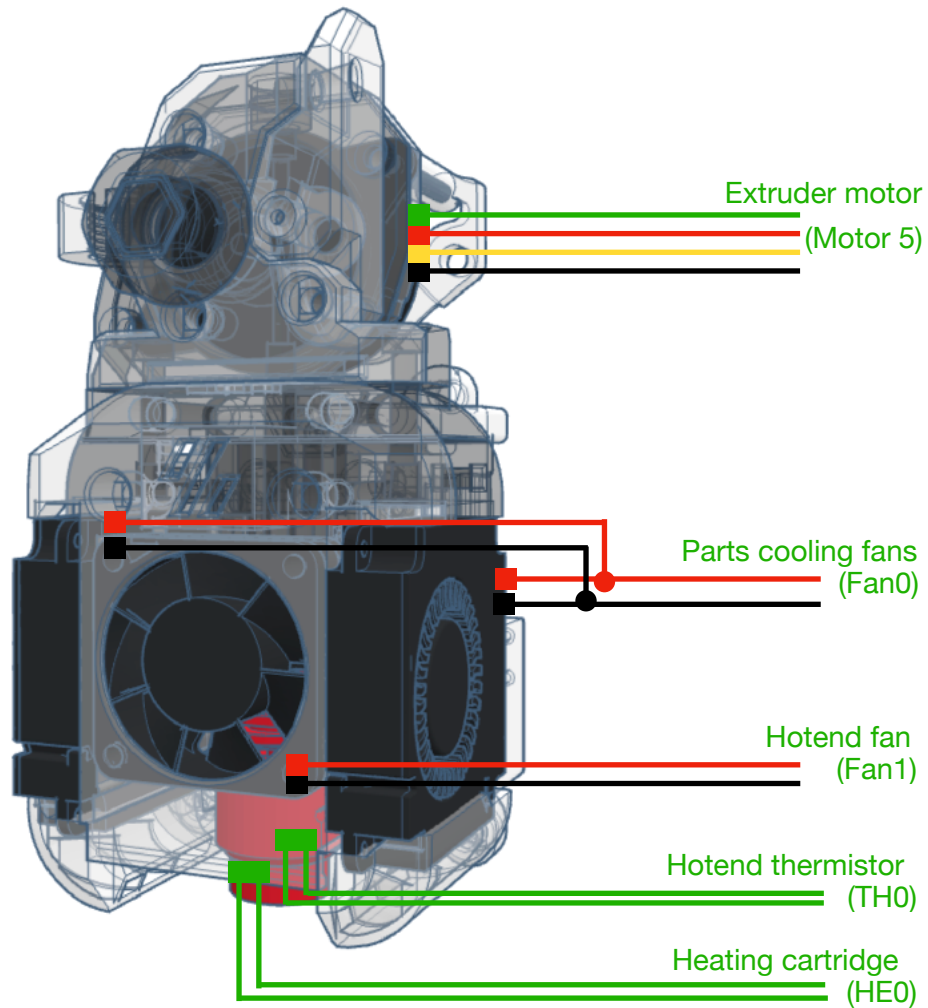
49



Mainboard installation

Toolhead and heatbed wires

50

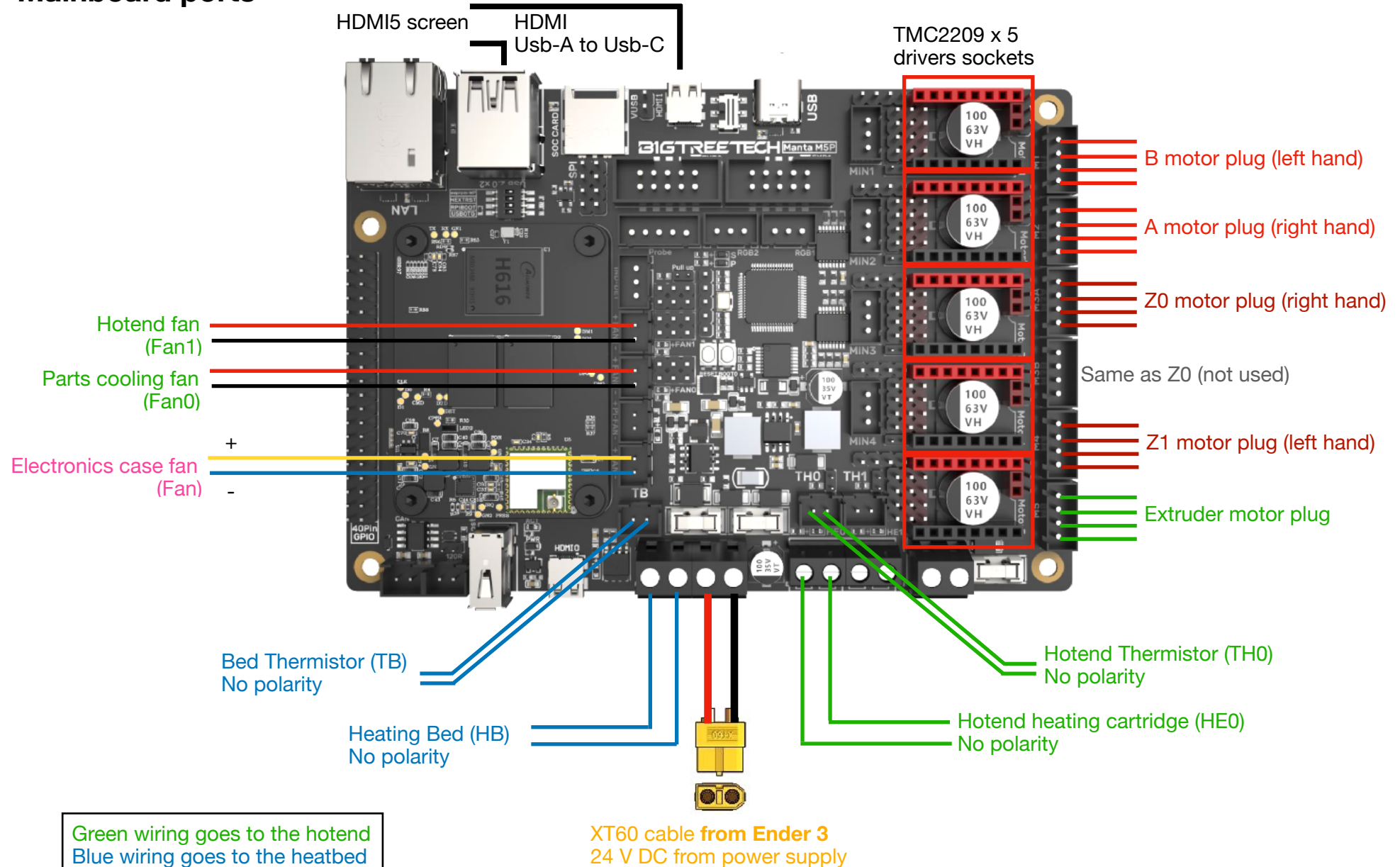


As an example, a Dragonburner toolhead with a Galileo2 extruder and a E3D Voron hotend
However, most toolhead configurations will use the exact same connector types.

Mainboard installation

Mainboard ports

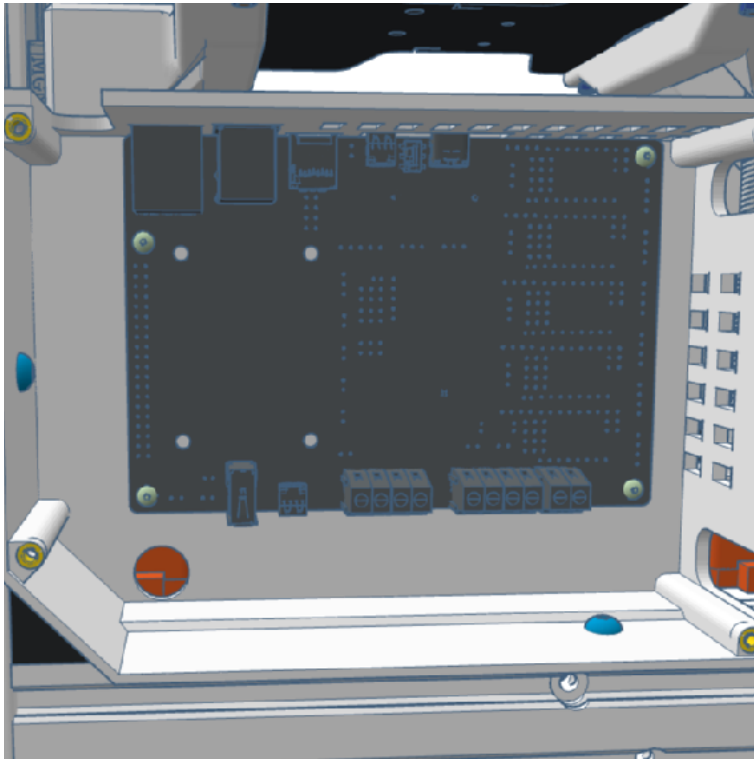
51



Mainboard installation

Going home

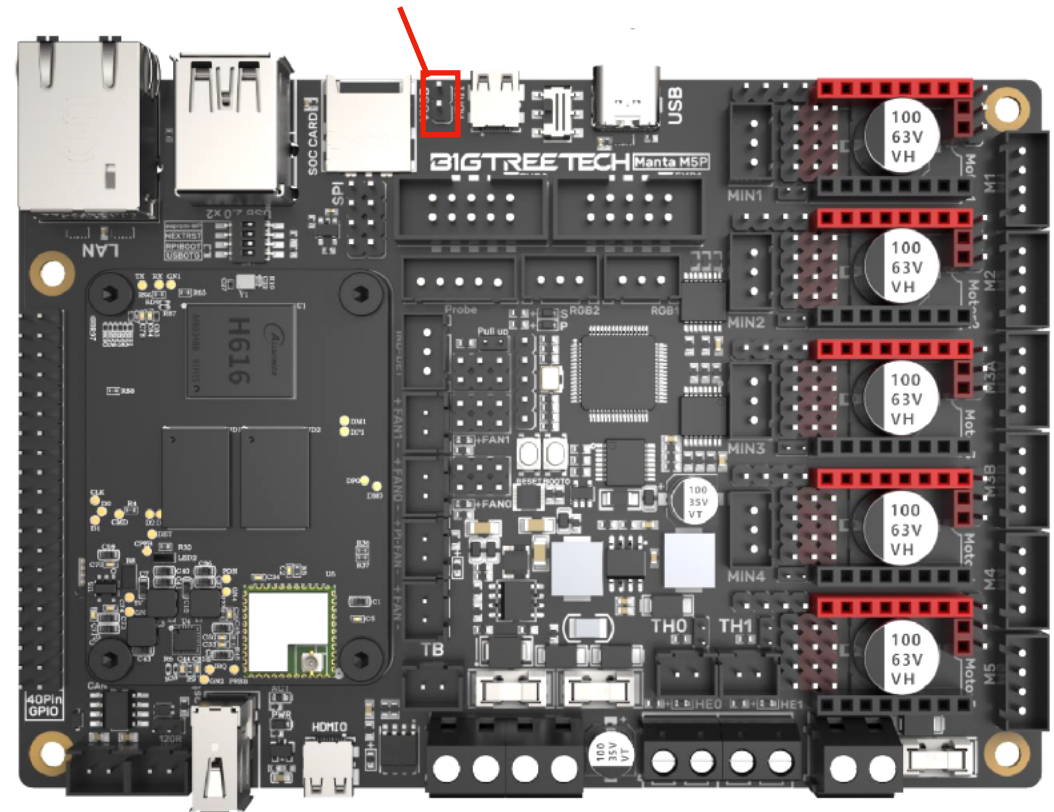
52



You should now be ready to insert your Manta 5P into the electronics control box
Secure it with 4 M3x6 BHCS screws, and go ahead with all your connections

Make good use of all the box cutouts to manage your cables and secure them with zipties or other sleeving of your choice

Temporary jumper to power the board by USB



To flash the image onto the CB1 microSD card, you may need to use the jumper shown here and power the mainboard with 5V by USB cable

Mainboard installation

53

(Nearly) Everything you ever wanted to know about Klipper



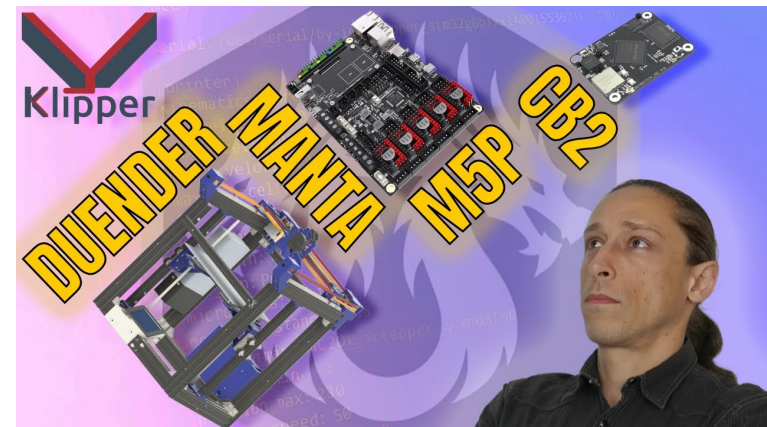
<https://2cm.es/1n0Yt>



<https://2cm.es/1i1Yq>



<https://2cm.es/1i1XM>

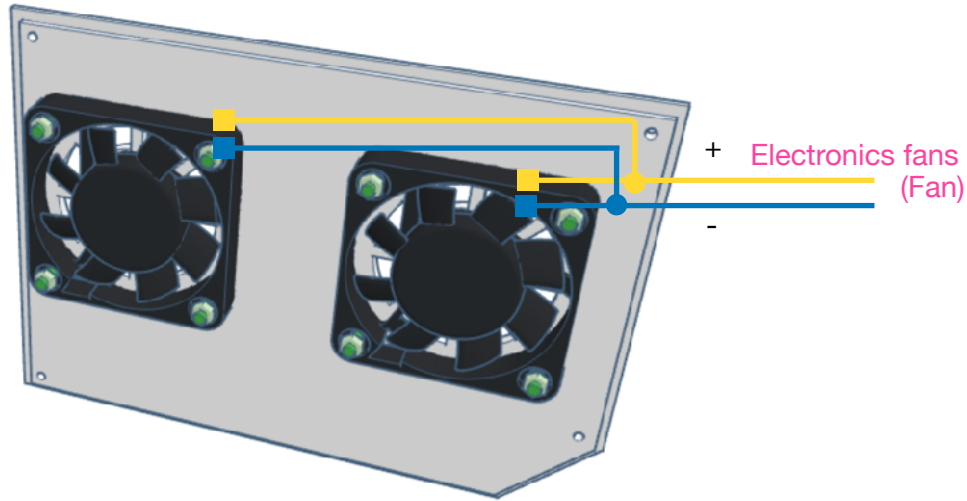


<https://2cm.es/1n0Wo>

Mainboard installation

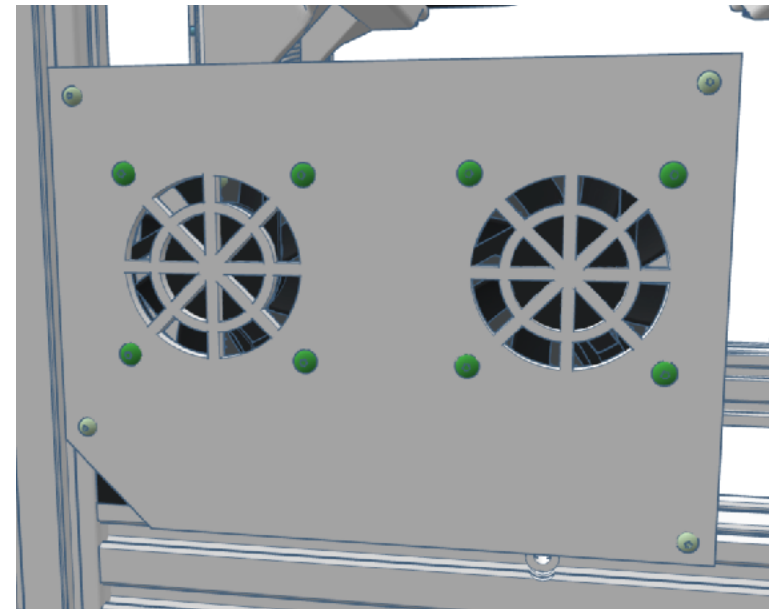
Electronics case fans

54



You may want to cool down your electronics with two 4010 axial fans taken **from the Ender 3 toolheads**

You can use 8 M3x18 BHCS screws with 8 M3 nuts to secure them to the box lid



Once you are all done, use the last 4 M3x6 BHCS screws to close the electronics box

Congratulations! Your build is now complete! Next step is installing Klipper and building your printer.cfg file
Once your printer is alive, we can only recommend the excellent [Ellis' Print Tuning Guide](https://ellis3dp.com/Print-Tuning-Guide/): <https://ellis3dp.com/Print-Tuning-Guide/>

We wish you loads of prints and fun with your Duender!

Coming soon!

A few words about Medusa HC project

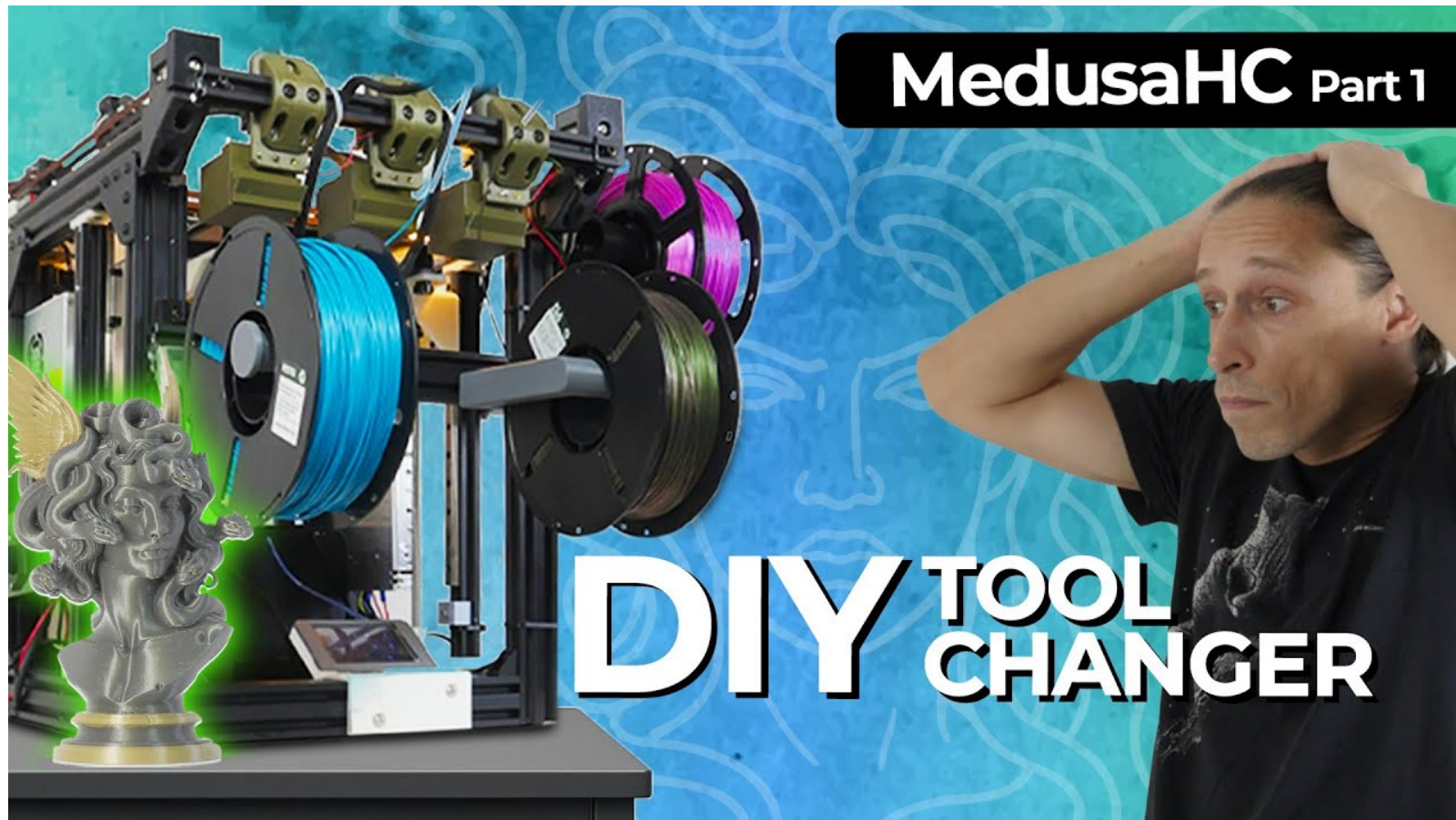
What? You want **more**?

Ok... What if I told you that your Duender can have **up to 4 different toolheads**?

What if you could print 4 different colours or experiment with 4 different materials at the same time, without needing to poop-purge?

Sounds great, doesn't it? Well stay tuned, because your **open-source, budget multi-toolhead mod for the Duender** is coming to you.

More info on Irbis 3D channel : <https://2cm.es/1h7sX>



<https://youtu.be/hpV5Z1TnGdY?si=FyJBHvE2mylvz7cY> or <https://2cm.es/1ml62>